Choice by Design:

Evidence from Feeding America's Food Allocation Problem

Sam M Altmann
Queen Mary University of London

April 2, 2026

Motivation

- Many organisations must allocate heterogeneous objects that arrive stochastically
  - Council houses to tenants
  - Donor kidneys to transplant patients
  - Contracts to contractors
  - Food to food banks

  $\rightarrow$ A central question is how much **Choice** should agents have?

Motivation

- Many organisations must allocate heterogeneous objects that arrive stochastically
  - Council houses to tenants
  - Donor kidneys to transplant patients
  - Contracts to contractors
  - Food to food banks

  $\rightarrow$ A central question is how much **Choice** should agents have?

- In this paper I study the allocation of food to food banks:
  - Food bank networks receive truckloads of various types of food and must decide which food bank to send it to
  - How important is it to allow food banks to choose the types of food they receive?

Motivation

- Many organisations must allocate heterogeneous objects that arrive stochastically
  - Council houses to tenants
  - Donor kidneys to transplant patients
  - Contracts to contractors
  - Food to food banks
  - → A central question is how much **Choice** should agents have?

- In this paper I study the allocation of food to food banks:
  - Food bank networks receive truckloads of various types of food and must decide which food bank to send it to
  - How important is it to allow food banks to choose the types of food they receive?

- The benefits of choice depend on **heterogeneity** in match values
  - → Estimating the degree of heterogeneity is key to welfare analysis

Feeding America

FEEDING
AMERICA

Feeding America

Feeding America

Feeding America

- Food banks have become an important part of U.S. society
  - 1 in 7 Americans do not have consistent access to enough food to live a healthy life
  - Yet, 30-40% of all the food produced in the U.S. gets wasted

  → Feeding America's food banks distribute around 700,000 meals each day

Feeding America

- Food banks have become an important part of U.S. society
  - 1 in 7 Americans do not have consistent access to enough food to live a healthy life
  - Yet, 30-40% of all the food produced in the U.S. gets wasted

  → Feeding America's food banks distribute around 700,000 meals each day

- Before 2005 food banks were offered food at random
  - Different **food banks** want different **types of food** at different **points in time**
  - What they want is determined by what they do not receive from local donors

Feeding America

- Food banks have become an important part of U.S. society
  - 1 in 7 Americans do not have consistent access to enough food to live a healthy life
  - Yet, 30-40% of all the food produced in the U.S. gets wasted

  → Feeding America's food banks distribute around 700,000 meals each day

- After 2005 Feeding America introduced an Auction System
  - Food banks place bids on food using fake money
  - Food banks in areas with more poverty get more fake money

Feeding America

- Food banks have become an important part of U.S. society
  - 1 in 7 Americans do not have consistent access to enough food to live a healthy life
  - Yet, 30-40% of all the food produced in the U.S. gets wasted

  → Feeding America's food banks distribute around 700,000 meals each day

- After 2005 Feeding America introduced an Auction System
  - Food banks place bids on food using fake money
  - Food banks in areas with more poverty get more fake money

- **Research Question:** How does welfare compare under the two systems?

  → Who are the winners and losers?

  → What factors drive these welfare benefits?

Research Strategy:

- **Empirical strategy**:
  - Estimate food banks' demand functions, and how these vary over time
  - Counterfactuals: Compare allocations under the current and previous regimes

Research Strategy:

- **Empirical strategy**:
  - Estimate food banks' demand functions, and how these vary over time
  - Counterfactuals: Compare allocations under the current and previous regimes

- **Challenges**:
  1. Strategic bidding: I only observe food banks' bids, not their underlying values
  2. Forward looking bidders: Evidence of inter-temporal substitution
  3. Unobserved state: I do not observe stocks, a key determinant of demand

Research Strategy:

- **Empirical strategy**:
  - Estimate food banks' demand functions, and how these vary over time
  - Counterfactuals: Compare allocations under the current and previous regimes
- **Challenges**:
  1. Strategic bidding: I only observe food banks' bids, not their underlying values
  2. Forward looking bidders: Evidence of inter-temporal substitution
  3. Unobserved state: I do not observe stocks, a key determinant of demand
- **Solution:**
  - Estimate a dynamic multi-object auction model (solves 1 & 2)
  - Use a Bayesian approach to numerically integrate over unobserved stocks (solves 3)
    $\rightarrow$ "data augmentation": Infer changes in stocks from changes in bids and **winnings**

Preview of Findings

- Preferences and Heterogeneity:
  1. Large heterogeneity across food banks and across different types of food
  2. Evidence of heterogeneity within food banks over time due to variation in stocks

- The Importance of Choice:
  1. Equivalent to a 36% increase in the supply of food, or 84 tons each day
  2. 89% of food banks are estimated to be better off under the Auction System

Preview of Findings

- Preferences and Heterogeneity:
  1. Large heterogeneity across food banks and across different types of food
  2. Evidence of heterogeneity within food banks over time due to variation in stocks
- The Importance of Choice:
  1. Equivalent to a 36% increase in the supply of food, or 84 tons each day
  2. 89% of food banks are estimated to be better off under the Auction System
- The key reason? Batching
  - The Auction System allocates many loads simultaneously in batches
  - The Old System allocated sequentially - each load allocated before the next arrived

    → Most other food bank networks allocate food sequentially

This paper makes two main contributions:

Contribution and Related Literature

This paper makes two main contributions:

1. Empirical: I examine how food banks benefit from choice over their allocations
   - **Empirical Market Design**: [*school choice*] Agarwal and Somaini (2018) Kapor et al (2020), Akbarpour et al (2022) [*teacher allocation*] Combe et al (2022), Bobba et al (2022) [*medical match*] Agarwal (2015), [*spectrum allocation*] Fox & Bajari (2013)
   - **Empirical Dynamic Matching**: [*public housing*] Thakral (2016), Waldinger (2022), van Dijk (2022), [*kidney allocation*] Agarwal et al (2020), Agarwal et al (2021), [*hunting permits*] Reeling & Verdier (2022), [*foster care*] Robinson-Cortés (2019)
   - **Food banks**: Prendergast (2017 & 2022), Walsh (2015), Lee et al (2025)

   → I use a long panel of food bank bidding data to estimate a structural model with rich, unobserved, and time varying, heterogeneity

Contribution and Related Literature

This paper makes two main contributions:

1. Empirical: I examine how food banks benefit from choice over their allocations
   - **Empirical Market Design**: [*school choice*] Agarwal and Somaini (2018) Kapor et al (2020), Akbarpour et al (2022) [*teacher allocation*] Combe et al (2022), Bobba et al (2022) [*medical match*] Agarwal (2015), [*spectrum allocation*] Fox & Bajari (2013)
   - **Empirical Dynamic Matching**: [*public housing*] Thakral (2016), Waldinger (2022), van Dijk (2022), [*kidney allocation*] Agarwal et al (2020), Agarwal et al (2021), [*hunting permits*] Reeling & Verdier (2022), [*foster care*] Robinson-Cortés (2019)
   - **Food banks**: Prendergast (2017 & 2022), Walsh (2015), Lee et al (2025)

   → I use a long panel of food bank bidding data to estimate a structural model with rich, unobserved, and time varying, heterogeneity

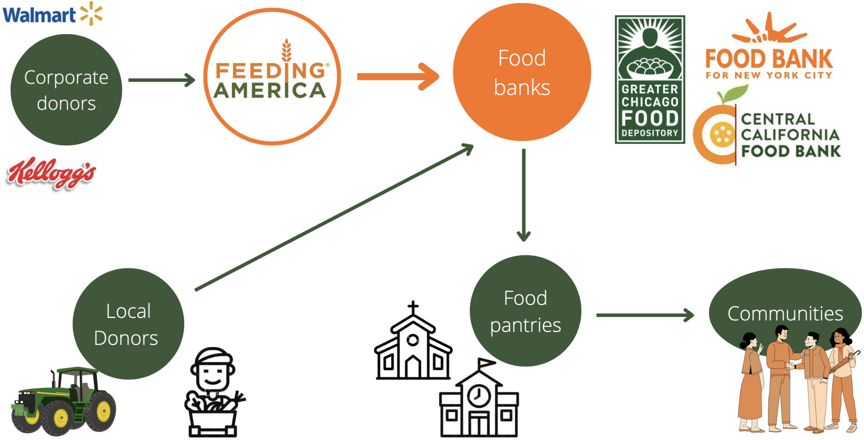
2. Methodological: I estimate an empirical auction model with unobserved states
   - **Empirical Auctions**: [*single object*] Guerre et al (2000) [*dynamic auctions*] Jofre-Bonet & Pesendorfer (2003), Backus & Lewis (2023), [*multi-object auctions*] Cantillon & Pesendorfer (2006), Gentry et al (2022) [*both*] Altmann (WP)
   - **Identification**: Berry & Compiani (2022), Connault (2016), Hu & Shum (2012), Arcidiacono & Miller (2011), Kasahara & Shimotsu (2009), Hendel & Nevo (2006)

   → I present an identification strategy using observed shifters of the unobserved states

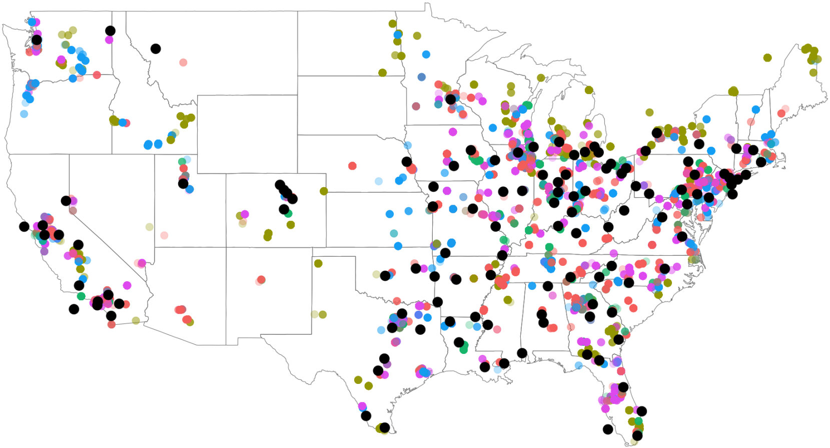
Outline

1. Institutions and Data
2. Model and Identification
3. Estimation and Results
4. Counterfactuals

Feeding America



Feeding America



Storage type: dried fresh nf refrigerated tinned

The Choice System (2005 - present):

- Around 300 tons of food is allocated each day
  - Each load is around 10-20 tons
  - Food banks organise transport themselves
  - Excludes Fresh Produce, which is allocated separately
- Loads are allocated using simultaneous First Price Sealed Bid auctions
  - i.e. They cannot place combination bids
  - $\approx$ 30 lots allocated simultaneously each day

The Auction System

The Choice System (2005 - present):

- Around 300 tons of food is allocated each day
  - Each load is around 10-20 tons
  - Food banks organise transport themselves
  - Excludes Fresh Produce, which is allocated separately
- Loads are allocated using simultaneous First Price Sealed Bid auctions
  - i.e. They cannot place combination bids
  - $\approx$ 30 lots allocated simultaneously each day
- Food banks place bids on loads using a fake currency - ‘shares’
  - Daily allocations of fake money are determined by local poverty
  - fake money can be saved, and interest free credit is available
  - Negative bids are allowed, down to -2000 $\rightarrow$ this helps shift undesirable loads
  - The (fake) money supply varies with the food supply to keep prices constant

Data

Two sources of data are used:

1. Bidding Data
   - Information on every auction from 2014-2017
   - The goods included in each lot
   - The location of each lot
   - Identities and bids of both winning and losing bidders

2. Food bank data
   - Catchment areas, local population data and poverty rates

→ I do not have data on stocks, local donations, or food sent to food pantries

The Auction System

Lots	Mean	SD
Daily lots	32.74	16.8
Tons per lot	13.25	5.45
Winning bid	2134	5818
Bids per lot	2.8	3.21
Bids per day	68.8	31.2
% Allocated	91	-
% Negative	34	-

Food Banks	Mean	p25	p50	p75
Population (000s)	1913	676	1270	2543
Poverty (000s)	284	99	191	373
Goal Factor	1	0.36	0.62	1.19
Bids per day	0.43	0.04	0.16	0.47
Average Bid	4140	1322	2760	4684
Wins per day	0.16	0.02	0.07	0.18
Average Payment	4571	1332	3033	5881

Note: Winning bids includes the reserve if no bids are received, hence smaller than average payment. 'Allocated' gives the percentage of lots receiving at least one bid, does not include lots allocated the following day. Negative prices include loads allocated for 0 shares. Statistics are calculated by food bank, then quantiles evaluated across food banks. The mean Goal Factor is normalised to 1. Population and Poverty figures refer to the people in a food bank's catchment area.

Outline

The model needs to incorporate 3 key observed facts:

1. Negative prices and infrequent bidding Graphs
   - 21% of bids are negative, and the average bidder only bids on 2% of lots

     $\rightarrow$ Evidence of **storage costs**

Stylised Facts

The model needs to incorporate 3 key observed facts:

1. Negative prices and infrequent bidding [Graphs]
   - 21% of bids are negative, and the average bidder only bids on 2% of lots

     → Evidence of **storage costs**

2. Inter-temporal substitution [Graphs]
   - After winning a lot, the probability of bidding on a similar lot falls by 33%
   - Anecdotal evidence that food banks are forward looking and patient

     → Evidence we need a **Dynamic** auction framework

Stylised Facts

The model needs to incorporate 3 key observed facts:

1. Negative prices and infrequent bidding Graphs
   - 21% of bids are negative, and the average bidder only bids on 2% of lots

     → Evidence of **storage costs**

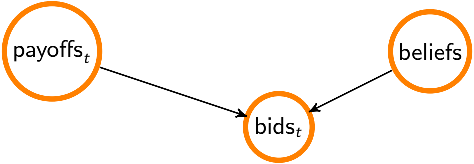
2. Inter-temporal substitution Graphs
   - After winning a lot, the probability of bidding on a similar lot falls by 33%
   - Anecdotal evidence that food banks are forward looking and patient

     → Evidence we need a **Dynamic** auction framework

3. Heterogeneity Food Banks Time
   - Systematic differences in behaviour across food banks and over time

     → Evidence of persistent and time-varying **unobserved heterogeneity**

The Model



- Food banks bid in repeated rounds of simultaneous first price auctions

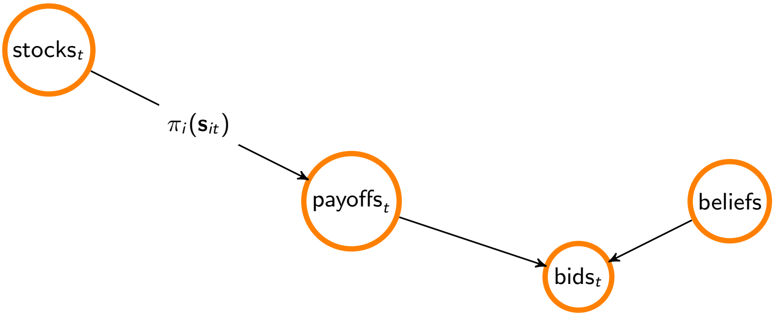
  Model details

  $\rightarrow$ Independent private values, endogenous entry into auctions, risk neutral bidders

  $\rightarrow$ Payoffs are Quasi-linear in fake money

  Equivalence Graphs

The Model

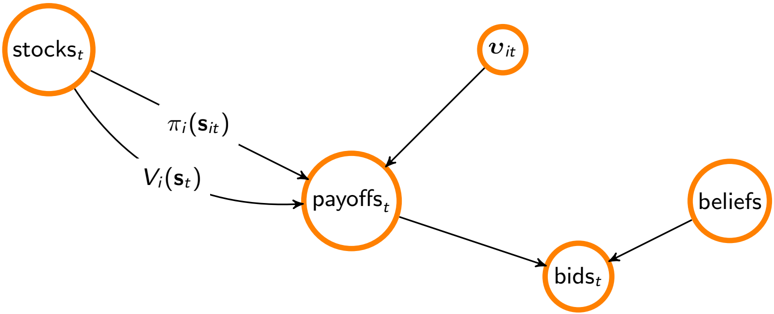


- If a food bank ends the period with stocks $\mathbf{s}_i$, they receive pay-off $\pi_i(\mathbf{s}_i)$

Model details

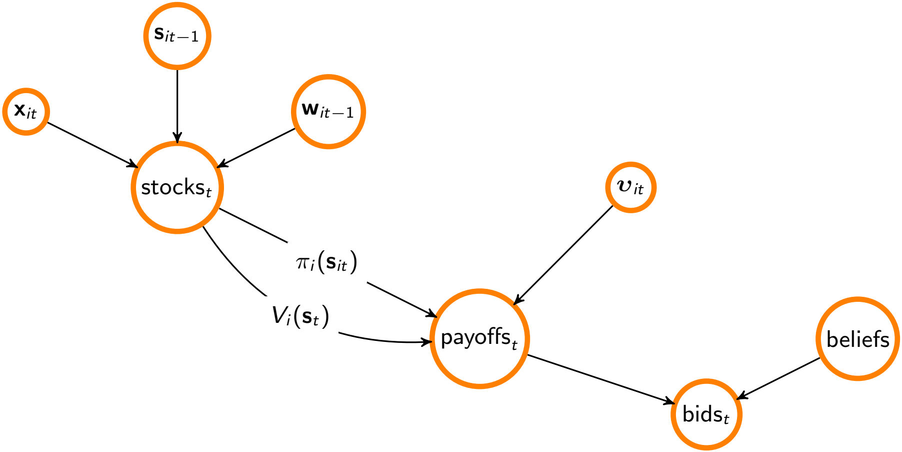
  $\rightarrow$ This depends on stock by subcategory and by storage type

  $\rightarrow$ This captures the utility of holding food to give it out, and storage costs



- $V_i(\mathbf{s})$ gives the continuation value: expected future payoffs given ending in state $\mathbf{s}$
- If they win lot l they also receive lot specific idiosyncratic pay-off $\upsilon_{itl} \sim F^{\upsilon}$
  $\rightarrow$ This captures transportation costs and unmodelled variation in lot attributes

The Model

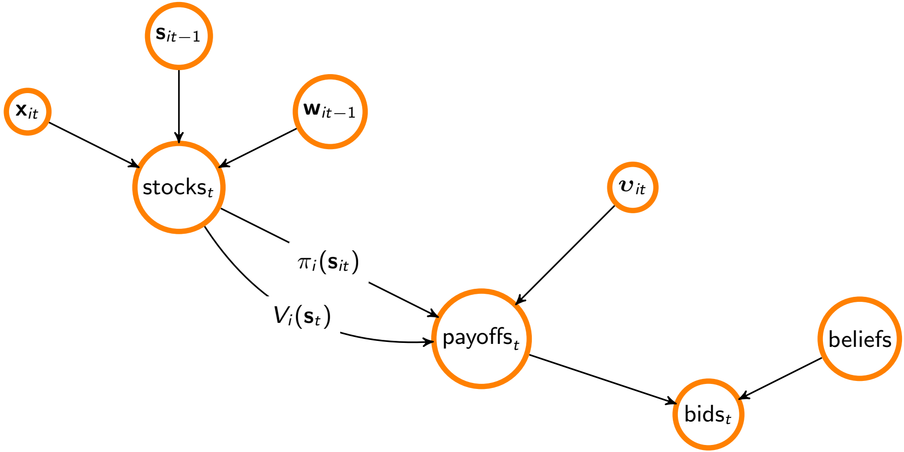


- I assume a stock process: $\mathbf{s}_{it} = \mathbf{s}_{it-1} + winnings_{it-1} + \mathbf{x}_{it}$

$\rightarrow$ $\mathbf{x}_{it}$ = local donations minus food distributed to local pantries $\mathbf{x}_{it} \sim F_i^x(\mathbf{s}_{it-1})$

$\rightarrow$ This is not a random walk: $\mathbf{w}_{it-1}$ and $\mathbf{x}_{it}$ respond to $\mathbf{s}_{it-1}$ to prevent stocks falling low

The Model

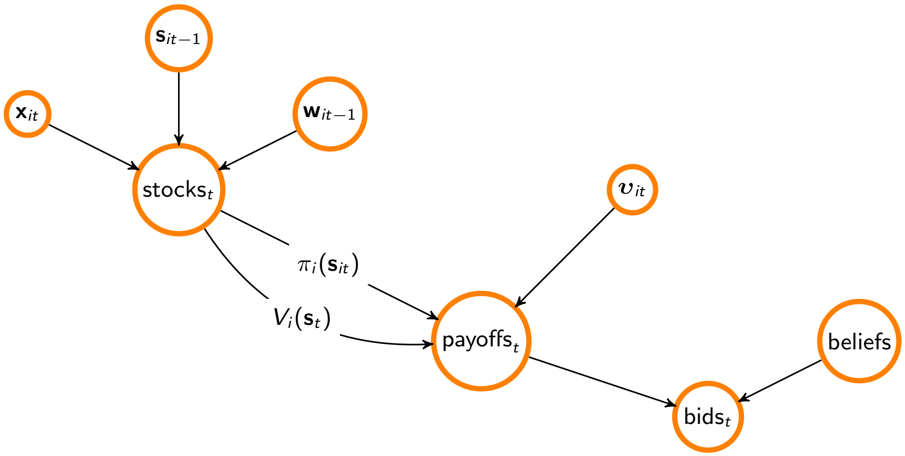


- I assume a stock process: $\mathbf{s}_{it} = \mathbf{s}_{it-1} + winnings_{it-1} + \mathbf{x}_{it}$

$\rightarrow$ $\mathbf{x}_{it}$ = local donations minus food distributed to local pantries $\mathbf{x}_{it} \sim N(\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}, \Sigma_i)$

$\rightarrow$ This is not a random walk: $\mathbf{w}_{it-1}$ and $\mathbf{x}_{it}$ respond to $\mathbf{s}_{it-1}$ to prevent stocks falling low

The Model



- **Equilibrium**: I focus on Markov Perfect Equilibrium Details
- I also make a Large Market Assumption on the equilibrium distribution of winning bids:
  → Prices may vary with common states (e.g. supply), but not with individual stocks $\mathbf{s}_{jt}$

The Food Bank's Problem

Sum over possible outcomes a

Bellman Equation:

$$W_i(\boldsymbol{\upsilon}, \mathbf{s}) = \max_{\mathbf{b}} \left\{ \textstyle\sum_l \Gamma_l(b_l)(\upsilon_l - b_l) + \textstyle\sum_a P_a(\mathbf{b})[\pi_i(\mathbf{s}_i^a) + \beta V_i(\mathbf{s}^a)] \right\}$$

$P(\text{win lot } l | b_l)$

Combination probability

$= E_{\boldsymbol{\upsilon}_{t+1}, \mathbf{s}_{t+1}}[W_i(\boldsymbol{\upsilon}, \mathbf{s}) | \mathbf{s}^a]$

Key:

- b_l = bid on lot l
- $\mathbf{s}_i$ = i's stocks
- $\mathbf{s}_i^a$ = i's hypothetical stocks
- υ_l = lot specific value
- $\pi_i(\mathbf{s}_i)$ = flow payoffs
- $V_i(\mathbf{s}_i)$ = continuation value
- β = discount factor

The Food Bank's Problem

Sum over possible outcomes a

Bellman Equation:

$$W_i(\boldsymbol{\upsilon}, \mathbf{s}) = \max_{\mathbf{b}} \left\{ \sum_l \Gamma_l(b_l)(\upsilon_l - b_l) + \sum_a P_a(\mathbf{b})[\pi_i(\mathbf{s}_i^a) + \beta V_i(\mathbf{s}^a)] \right\}$$

$P(\text{win lot } l | b_l)$

Combination probability

$= E_{\boldsymbol{\upsilon}_{t+1}, \mathbf{s}_{t+1}}[W_i(\boldsymbol{\upsilon}, \mathbf{s}) | \mathbf{s}^a]$

- **First Order Conditions:**

$$0 = \frac{\partial \Gamma_l(b_l^*)}{\partial b_l}(\upsilon_l - b_l^*) - \Gamma_l(b_l^*) + \sum_a \frac{\partial P_a(\mathbf{b}^*)}{\partial b_l}[\pi_i(\mathbf{s}_i^a) + \beta V_i(\mathbf{s}^a)]$$

The Food Bank's Problem

Bellman Equation:

$$W_i(\boldsymbol{\upsilon}, \mathbf{s}) = \max_{\mathbf{b}} \left\{ \sum_l \underbrace{\Gamma_l}_{P(\text{win lot } l \mid b_l)}(b_l)(\upsilon_l - b_l) + \overbrace{\sum_a}^{\text{Sum over possible outcomes } a} \underbrace{P_a}_{\text{Combination probability}}(\mathbf{b})[\pi_i(\mathbf{s}_i^a) + \beta \underbrace{V_i(\mathbf{s}^a)}_{= E_{\boldsymbol{\upsilon}_{t+1}, \mathbf{s}_{t+1}}[W_i(\boldsymbol{\upsilon}, \mathbf{s}) \mid \mathbf{s}^a]}] \right\}$$

- **First Order Conditions:**

$$0 = \frac{\partial \Gamma_l(b_l^*)}{\partial b_l}(\upsilon_l - b_l^*) - \Gamma_l(b_l^*) + \sum_a \frac{\partial P_a(\mathbf{b}^*)}{\partial b_l}[\pi_i(\mathbf{s}_i^a) + \beta V_i(\mathbf{s}^a)]$$

- **Inverse Bid System:**

$$\upsilon_l(\mathbf{b}; \mathbf{s}) = b_l + \frac{\Gamma_l(b_l)}{\partial \Gamma_l(b_l)/\partial b_l} - \sum_a \frac{\partial P_a(\mathbf{b})/\partial b_l}{\partial \Gamma_l(b_l)/\partial b_l}[\pi_i(\mathbf{s}_i^a) + \beta V_i(\mathbf{s}^a)]$$

Identification

Are the model primitives $\{\pi_i(\mathbf{s}_i), F_i^x\}_i$ identified from our data?

- Identification is a major challenge, particularly due to...
  - $\rightarrow$ Unobserved states
  - $\rightarrow$ Simultaneous auctions, Repeated auctions, Reservation prices

Identification

Are the model primitives $\{\pi_i(\mathbf{s}_i), F_i^x\}_i$ identified from our data?

- Identification is a major challenge, particularly due to...
  - $\rightarrow$ Unobserved states
  - $\rightarrow$ Simultaneous auctions, Repeated auctions, Reservation prices $\}$ Altmann (WP)

Identification

Are the model primitives $\{\pi_i(\mathbf{s}_i), F_i^x\}_i$ identified from our data?

- Identification is a major challenge, particularly due to...
  - $\rightarrow$ Unobserved states
  - $\rightarrow$ Simultaneous auctions, Repeated auctions, Reservation prices $\}$ Altmann (WP)

- Building on Hu and Shum (2012) identification exploits 2 sources of variation:
  1. Using observed variation in the size and composition of lots Intuition
     - $\rightarrow$ This pins down $\pi_i(\mathbf{s}_i)$
  2. Using observed variation in winnings Intuition
     - $\rightarrow$ This pins down $F_i^{\mathbf{x}}$

Proof Outline

Outline

Estimation of Dynamic Auction Models

- The standard estimation procedure comes from Jofre-Bonet & Pesendorfer (2003)
  1. Estimate the distribution of equilibrium bids
  2. Write $V(\mathbf{s})$ as a function of this distribution
  3. Back out $\pi(\mathbf{s})$

     $\rightarrow$ This relies on a 1:1 equilibrium mapping from types to bids

Skip

Abridged

Estimation of Dynamic Auction Models

- The standard estimation procedure comes from Jofre-Bonet & Pesendorfer (2003)
  1. Estimate the distribution of equilibrium bids
  2. Write $V(\mathbf{s})$ as a function of this distribution
  3. Back out $\pi(\mathbf{s})$

     $\rightarrow$ This relies on a 1:1 equilibrium mapping from types to bids

Skip

Abridged

- This is not possible in a Multi-object environment

  $\rightarrow$ We cannot write V as a function of the distribution of bids **only**

Estimation of Dynamic Auction Models

- The standard estimation procedure comes from Jofre-Bonet & Pesendorfer (2003)
  1. Estimate the distribution of equilibrium bids
  2. Write $V(\mathbf{s})$ as a function of this distribution
  3. Back out $\pi(\mathbf{s})$

     $\rightarrow$ This relies on a 1:1 equilibrium mapping from types to bids

Skip

Abridged

- This is not possible in a Multi-object environment

  $\rightarrow$ We cannot write V as a function of the distribution of bids **only**

- Altmann (WP): we *can* write V as a function of bids and a correction term

  $\rightarrow$ This correction is just a (linear) function of $k_i(\mathbf{s}) = \pi_i(\mathbf{s}_i) + \beta V(\mathbf{s})$

  [The Pseudo-Primitives of a 'misspecified' static game]

  $\rightarrow$ If we 'knew' $k_i(\mathbf{s})$ (and F_i^x), we could evaluate $V(\mathbf{s})$, then back out $\pi_i = k_i - \beta V_i$

3 Step Estimation Procedure

1. Estimate equilibrium beliefs: Γ & P

   Details

   - Winning bids are Generalised Extreme Value: $\max_j \{b_{jlt}\} \sim GEV(\alpha_l, \sigma_l, \mu_l + \nu_l(\mathbf{s}))$
   - Prices vary with *common* state variables (e.g. supply of similar food): $\nu_l(\mathbf{s}) = \mathbf{s}_t^{0T} \nu_l$

3 Step Estimation Procedure

1. Estimate equilibrium beliefs: Γ & P Details
   - Winning bids are Generalised Extreme Value: $\max_j\{b_{jlt}\} \sim GEV(\alpha_l, \sigma_l, \mu_l + \nu_l(\mathbf{s}))$
   - Prices vary with *common* state variables (e.g. supply of similar food): $\nu_l(\mathbf{s}) = \mathbf{s}_t^{0T}\nu_l$

2. Estimate F_i^x, F_i^υ, and $k_i(\mathbf{s}) = \pi_i(\mathbf{s}_i) + \beta V_i(\mathbf{s})$ (Details shortly)

   $\rightarrow$ This uses the Transition Process as well as the Inverse Bid Function:

$$\upsilon_l(\mathbf{b};\mathbf{s}) = b_l + \frac{\hat{\Gamma}_l(b_l)}{\partial\hat{\Gamma}_l(b_l)/\partial b_l} - \sum_a \frac{\partial\hat{P}_a(\mathbf{b})/\partial b_l}{\partial\hat{\Gamma}_l(b_l)/\partial b_l}[\pi_i(\mathbf{s}_i^a) + \beta V_i(\mathbf{s}^a)]$$

3 Step Estimation Procedure

1. Estimate equilibrium beliefs: Γ & P Details
   - Winning bids are Generalised Extreme Value: $\max_j\{b_{jlt}\} \sim GEV(\alpha_l, \sigma_l, \mu_l + \nu_l(\mathbf{s}))$
   - Prices vary with *common* state variables (e.g. supply of similar food): $\nu_l(\mathbf{s}) = \mathbf{s}_t^{0\,T}\nu_l$

2. Estimate F_i^x, F_i^υ, and $k_i(\mathbf{s}) = \pi_i(\mathbf{s}_i) + \beta V_i(\mathbf{s})$ (Details shortly)

   $\rightarrow$ This uses the Transition Process as well as the Inverse Bid Function:

$$\upsilon_l(\mathbf{b};\mathbf{s}) = b_l + \frac{\hat{\Gamma}_l(b_l)}{\partial\hat{\Gamma}_l(b_l)/\partial b_l} - \sum_a \frac{\partial\hat{P}_a(\mathbf{b})/\partial b_l}{\partial\hat{\Gamma}_l(b_l)/\partial b_l}[\pi_i(\mathbf{s}_i^a) + \beta V_i(\mathbf{s}^a)]$$

3. Disentangle π_i and V_i Details
   - Write $V_i(\mathbf{s})$ as a function of F_i^x and $k_i(\mathbf{s})$:

$$\hat{V}_i(\mathbf{s}) = E_{\upsilon,\mathbf{s}'}\left[\max_b\left\{\sum_l \hat{\Gamma}_l(b_l)(\upsilon_l - b_l) + \sum_a \hat{P}_a(\mathbf{b})[\widehat{\pi_i(\mathbf{s}_i^a) + \beta V_i(\mathbf{s}^a)}]\right\}|\mathbf{s}\right]$$

   - Then back out $\hat{\pi} = \hat{k} - \beta\hat{V}$

3 Step Estimation Procedure

1. Estimate equilibrium beliefs: Γ & P Details
   - Winning bids are Generalised Extreme Value: $\max_j\{b_{jlt}\} \sim GEV(\alpha_l, \sigma_l, \mu_l + \nu_l(\mathbf{s}))$
   - Prices vary with *common* state variables (e.g. supply of similar food): $\nu_l(\mathbf{s}) = \mathbf{s}_t^{0\,T}\nu_l$

2. Estimate F_i^x, F_i^υ, and $k_i(\mathbf{s}) = \pi_i(\mathbf{s}_i) + \beta V_i(\mathbf{s})$ (Details shortly)

   $\rightarrow$ This uses the Transition Process as well as the Inverse Bid Function:

$$\upsilon_l(\mathbf{b};\mathbf{s}) = b_l + \frac{\hat{\Gamma}_l(b_l)}{\partial\hat{\Gamma}_l(b_l)/\partial b_l} - \sum_a \frac{\partial \hat{P}_a(\mathbf{b})/\partial b_l}{\partial\hat{\Gamma}_l(b_l)/\partial b_l}[\pi_i(\mathbf{s}_i^a) + \beta V_i(\mathbf{s}^a)]$$

3. Disentangle π_i and V_i Details
   - Write $V_i(\mathbf{s})$ as a function of F_i^x and $k_i(\mathbf{s})$:

$$\hat{V}_i(\mathbf{s}) = E_{\mathbf{b},\mathbf{s}'}\left[\sum_l \frac{\hat{\Gamma}_l(b_l)^2}{\partial\hat{\Gamma}_l(b_l)/\partial b_l} + \sum_a[\hat{P}_a(\mathbf{b}) - \sum_l \hat{\Gamma}_l(b_l)\frac{\partial \hat{P}_a(\mathbf{b})/\partial b_l}{\partial\hat{\Gamma}_l(b_l)/\partial b_l}][\hat{k}_i(\mathbf{s}^a)]\ |\mathbf{s}\right]$$

   - Then back out $\hat{\pi} = \hat{k} - \beta\hat{V}$

Estimating F_i^x and $k_i(\mathbf{s})$

- I make parametric assumptions on F_i^x and $k_i(\mathbf{s})$:

Details

  - $\mathbf{x}_{it}$ is normally distributed, with mean $\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}$ and variance Σ_i
  - $k_i(\mathbf{s})$ is quadratic in $\mathbf{s}_i$ (Linear Demand Curves)

- The general idea behind estimation:
  1. Observe how behaviour changes with the available lots, or around a recent win
  2. Consider other changes in bidding behaviour...
     ...and find the change in stocks that rationalise this behaviour

Estimating F_i^x and $k_i(\mathbf{s})$

- I make parametric assumptions on F_i^x and $k_i(\mathbf{s})$: Details
  - $\mathbf{x}_{it}$ is normally distributed, with mean $\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}$ and variance Σ_i
  - $k_i(\mathbf{s})$ is quadratic in $\mathbf{s}_i$ (Linear Demand Curves)

- Estimated using a Gibbs Sampler: Details
  1. *k-step*: Given draw of $\{\mathbf{s}_t\}_T$, sample k

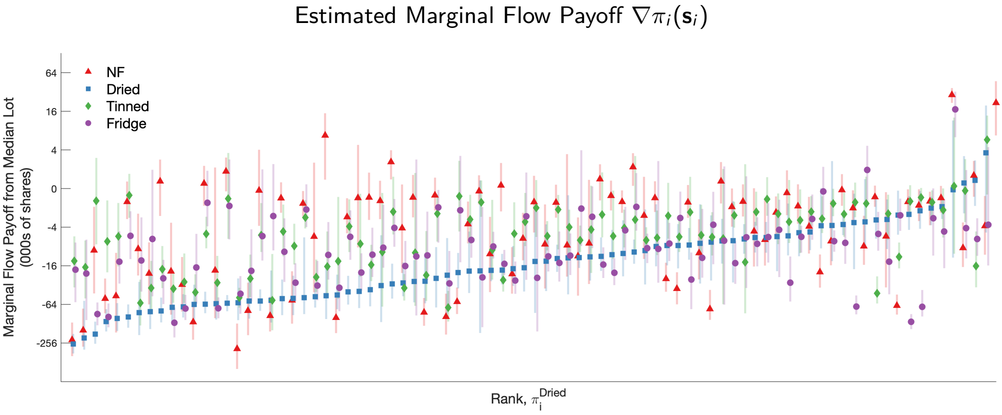
     $\rightarrow$ Regress available lots and $\mathbf{s}$ on bids

     $\rightarrow$ This relationship arises from the inverse bid system
  2. *s-step*: Given draw of k, sample $\{\mathbf{s}_t\}_T$ and hence F^x

     $\rightarrow$ Employs the Carter-Kohn Algorithm

     $\rightarrow$ Infer changes in stocks from changes in bids and winnings
  3. repeat

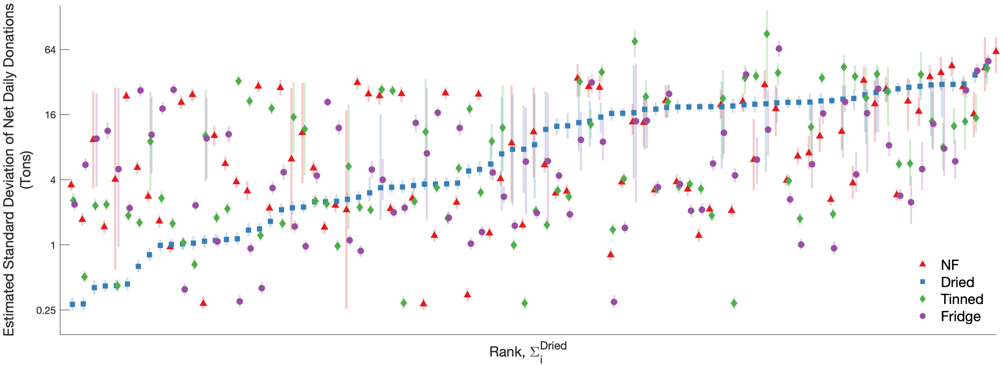
Results: Storage Costs



Note: Plot shows estimated marginal flow pay-off from receiving an average lot by food bank $\times$ food type, evaluated at their long run average stocks. Estimates are sorted across food banks by estimate for Dried food. 95% credible intervals are plotted. See $\hat{k}$ See $\hat{\Phi}$ See $\hat{\Phi}^h$ See $\hat{\lambda}$ See $\hat{dist}$ See Model Fit

Results: Net Donations

Estimated Standard Deviation of Net Donations ($\sqrt{\boldsymbol{\Sigma}_i} = \sqrt{\hat{Var}[\mathbf{x}_{it}]}$)



Note: Plot shows estimated standard deviation of net donations by food bank $\times$ food type, sorted across food banks by estimate for Dried food (blue). Error bars give 95% credible intervals. See $\hat{\mu}$ See $\hat{\delta}$ See $\hat{\sigma}$

Outline

1. Institutions and Data
2. Model and Identification
3. Estimation and Results
4. Counterfactuals
   - Mechanisms
   - Welfare

I consider 3 mechanisms:

1. The Auction System
2. The Old System
    - Food banks queue, get offered a load, then go to the back of the queue
    - Due to time constraints each load could only be offered to ≈ 10 food banks
3. No System (benchmark)

- For each mechanism I need to solve for the Markov Perfect Equilibrium

    $\rightarrow$ Find the fixed point between Accept/Reject decisions and beliefs

Mechanisms

I consider 3 mechanisms:

1. The Auction System
2. The Old System
    - Food banks queue, get offered a load, then go to the back of the queue
    - Due to time constraints each load could only be offered to $\approx$ 10 food banks
3. No System (benchmark)

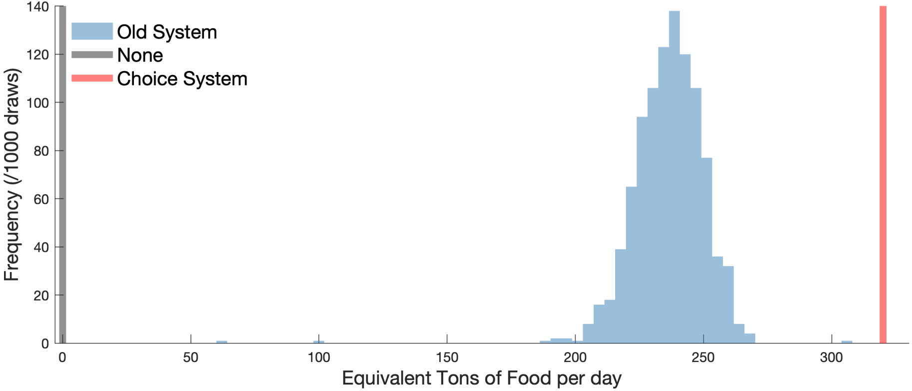
- For each mechanism I need to solve for the Markov Perfect Equilibrium

  $\rightarrow$ Find the fixed point between Accept/Reject decisions and beliefs

- Consider Welfare in terms of Consumer Surplus, measured in virtual currency
    - The money supply varies with the food supply to ensure prices remain constant

      $\rightarrow$ Hence we can translate welfare into equivalent increase in the food supply

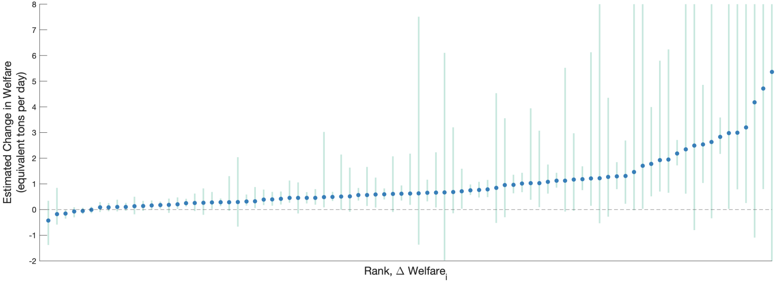
Welfare



Note: Plot shows the posterior distribution of welfare under each mechanism. Evaluated over 1000 draws from the posterior distribution of parameters. Welfare is measured relative to no allocation mechanism (level normalisation), and relative to the Auction System (scale normalisation), so that $\tilde{W}^{OS} = \frac{W^{OS} - W^{N}}{W^{AS} - W^{N}}$. On average, welfare increased by 84 tons of food per day, a gain of 36% relative to the Old System.

Equity?

- On average 89% of food banks achieve higher welfare under the Auction System



Note: Plot shows welfare by food bank, ordered by posterior mean, with 95% credible intervals.

- The food banks that lose out are the least picky

  → They hoover up the food rejected by other food banks

  → They tend to consume more, and lower quality, food under the Auction System

Where does this benefit come from?

- More food allocated [Histogram]

  → On average 34% more food is allocated under the Auction System

- Less distance travelled [Histogram]

  → On average lots are allocated 24% closer under the Auction System

- Food banks are less picky under the Old System

  → The average food bank is 3 times more likely to accept any given load

  → They accept food that doesn't exactly meet their needs in that moment

  → Food that a different food bank needs more

Welfare decomposition

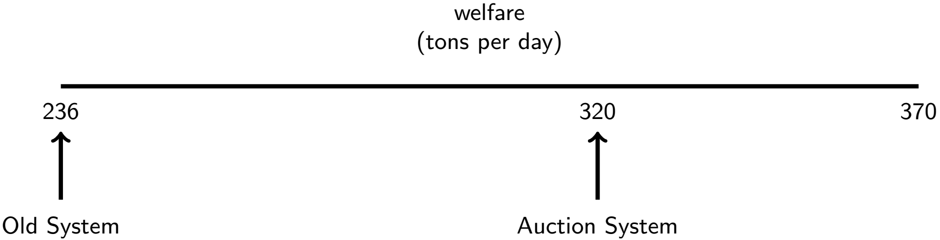
I consider several additional allocation mechanisms:

- Sequential Offers: work down the queue offering each load to **every** food bank
- Sequential Auctions: Infer the value of **signalling**
- Combinatorial Auctions: Decompose the **batching** and **exposure** effects

Welfare decomposition

I consider several additional allocation mechanisms:

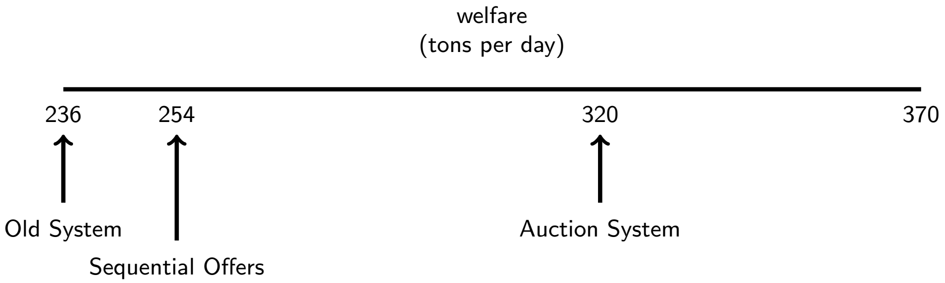
- Sequential Offers: work down the queue offering each load to **every** food bank
- Sequential Auctions: Infer the value of **signalling**
- Combinatorial Auctions: Decompose the **batching** and **exposure** effects



Welfare decomposition

I consider several additional allocation mechanisms:

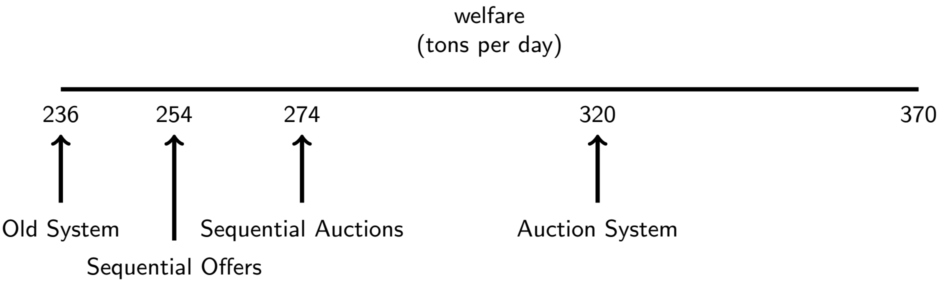
- Sequential Offers: work down the queue offering each load to **every** food bank
- Sequential Auctions: Infer the value of **signalling**
- Combinatorial Auctions: Decompose the **batching** and **exposure** effects



Welfare decomposition

I consider several additional allocation mechanisms:

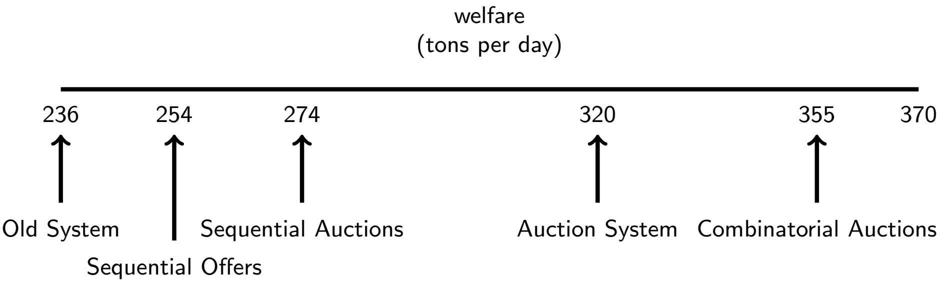
- Sequential Offers: work down the queue offering each load to **every** food bank
- Sequential Auctions: Infer the value of **signalling**
- Combinatorial Auctions: Decompose the **batching** and **exposure** effects



Welfare decomposition

I consider several additional allocation mechanisms:

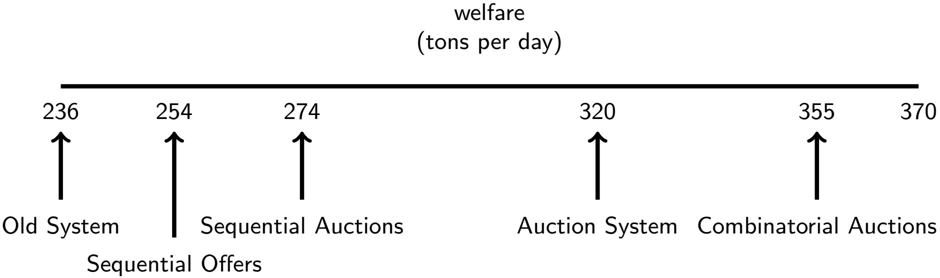
- Sequential Offers: work down the queue offering each load to **every** food bank
- Sequential Auctions: Infer the value of **signalling**
- Combinatorial Auctions: Decompose the **batching** and **exposure** effects



Welfare decomposition

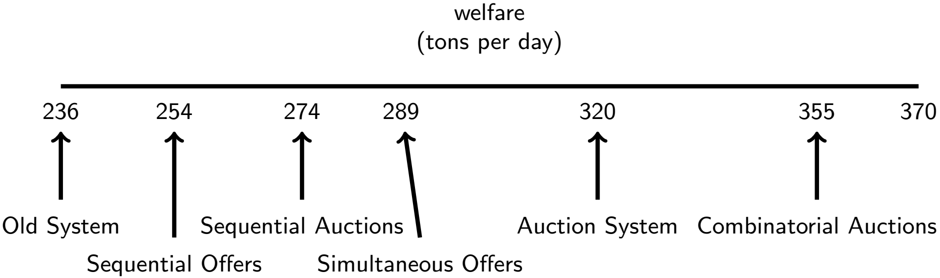
I consider several additional allocation mechanisms:

- Sequential Offers: work down the queue offering each load to **every** food bank
- Sequential Auctions: Infer the value of **signalling**
- Combinatorial Auctions: Decompose the **batching** and **exposure** effects
- Simultaneous Offers: A signal-free batched mechanism, offering Menus



I consider several additional allocation mechanisms:

- Sequential Offers: work down the queue offering each load to **every** food bank
- Sequential Auctions: Infer the value of **signalling**
- Combinatorial Auctions: Decompose the **batching** and **exposure** effects
- Simultaneous Offers: A signal-free batched mechanism, offering Menus



Summary

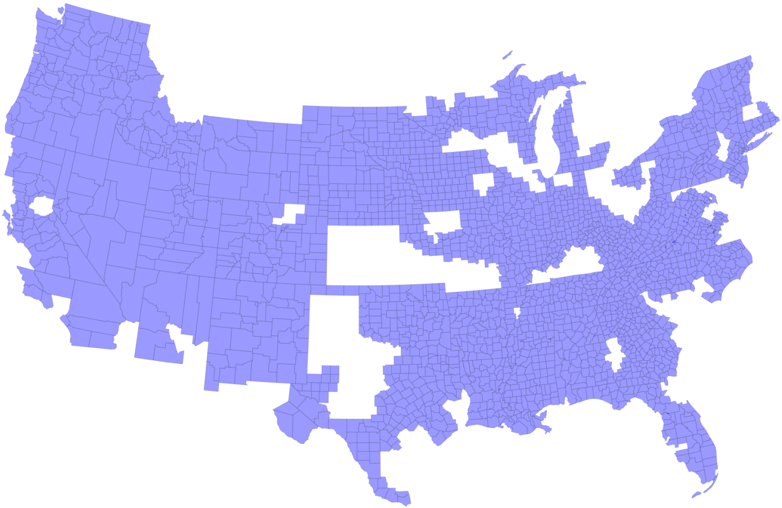
1. How important is it to give food banks Choice over the food they are allocated?
    - Applicable for numerous other food bank networks around the world
2. Developed a framework to estimate demand when stocks are unobserved
    - Found strong evidence of heterogeneity both across food banks and across time
3. Allowing Choice is extremely important
    - Increased welfare by an amount equivalent to increasing the food supply by 84 tons
    - This effect is driven by batching $\rightarrow$ larger choice sets $+$ better information

Summary

1. How important is it to give food banks Choice over the food they are allocated?
    - Applicable for numerous other food bank networks around the world
2. Developed a framework to estimate demand when stocks are unobserved
    - Found strong evidence of heterogeneity both across food banks and across time
3. Allowing Choice is extremely important
    - Increased welfare by an amount equivalent to increasing the food supply by 84 tons
    - This effect is driven by batching $\rightarrow$ larger choice sets + better information

Future directions:

- Is there room for improvement?
- How do these results translate to other food bank networks?

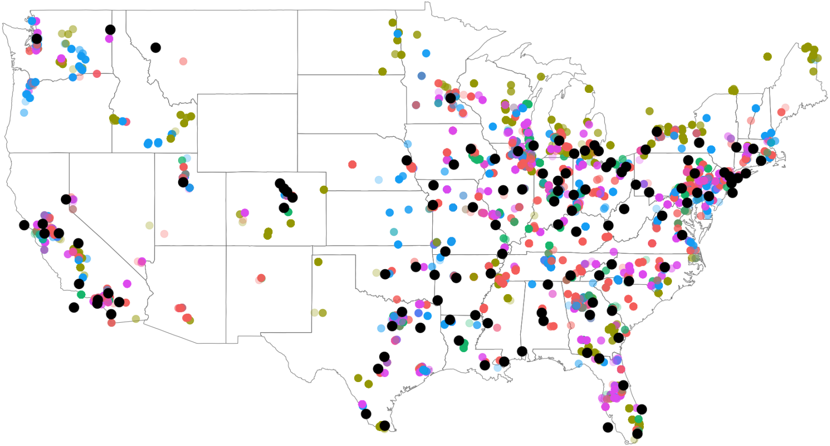


Note: Map plots counties in the catchment areas of food banks who make regular use of the Choice System.

return return


crackers & cookies
snacks
candy
condiments
non-dairy milk
bars
flavoured milk
coffee
dairy
lunch
pop
crisps
towel
dressing
cabbage
nappy
vegetables
soup
melon
corn flakes
oranges/lemons
squash
sugar
food
shake
yoghurt
milk
sauce
pasta
cookie
beans
peanut butter
detergent
onion
sausage
slim jims
pizza
meat
apple
breakfast
tomato
dessert
juice
water
potato
cheerio
carrot
cheese
meal
boxes
baked goods
Kellogg

return



Storage type: dried fresh nf refrigerated tinned

return return

Altmann (WP)

The Identification and Estimation of a Dynamic Multi-Object Auction Model

- Auctions rarely take place in isolation:
  - Many objects/contracts are often auctioned *simultaneously*
  - Auctions are *repeated* as new lots become available
  - Ignoring dynamics or complementarities will bias estimates

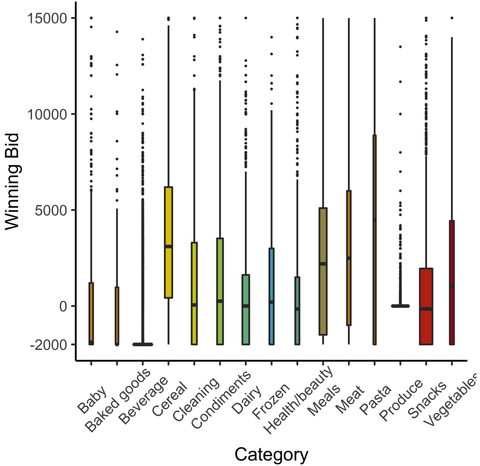
    $\rightarrow$ This is the first paper to unify the dynamic and multi-object setting

- I present a model of bidding in repeated rounds of simultaneous auctions
  - Identification / estimation is difficult because this is not a direct revelation mechanism
  - I show that primitives of the model are identified using observed variation in the state
  - I propose a computationally feasible estimation procedure (*more on this later!*)

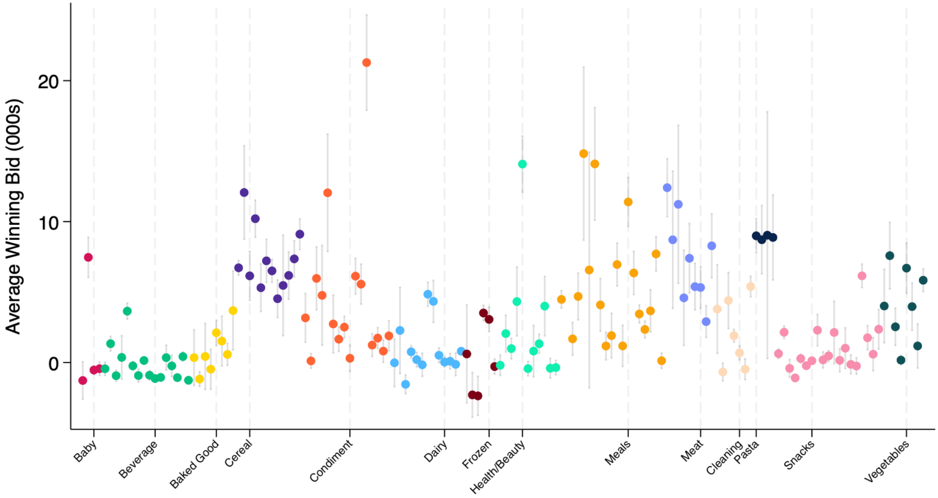
- I apply the model to data from Michigan's Department of Transport procurement

◂ literature ◂ estimation

Heterogeneity in food



- Strong evidence that some goods are preferred to others
- But lots of variation within categories



Note: Plot shows average winning bids across 164 subcategories of food. Controlling for size, location, and composition of the lot. Subcategories are divided within the 15 categories shown.

return

Reduced Form exercise:

Consider a simple Tobit regression:

- Split food into 5 types, according to how the food is stored:
  - $\rightarrow$ Dried, Tinned / Bottled, Refrigerated, Fresh, and Non-food
  - $\rightarrow$ This helps me focus on storage costs, and how they vary with stocks, as a key margin
- Find each food bank's average bid for each type of food

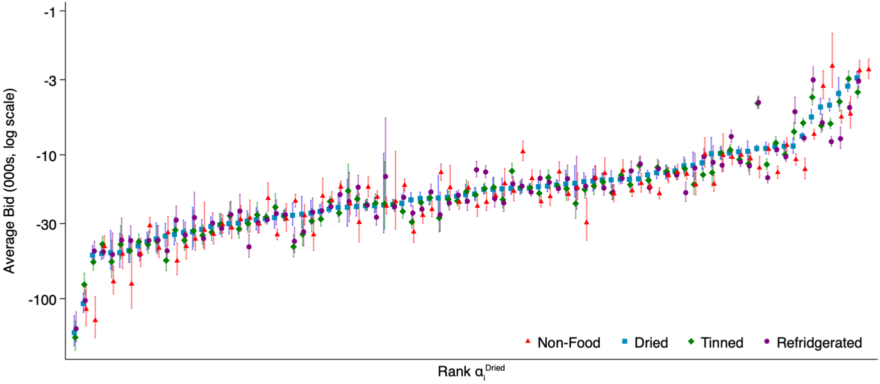
$$b_{itl} = \alpha_{ig} + \varepsilon_{itl} \qquad b^*_{itl} = \begin{cases} b_{itl} & \text{if} \quad b_{itl} \geq R_l \\ R_l & \text{if} \quad \text{Otheriwse} \end{cases}$$

$$\varepsilon_{itl} \sim N(0, \sigma_{il}) \tag{1}$$

- Where α_{ig} are food bank $\times$ type specific means

return

Heterogeneity Across Food Banks

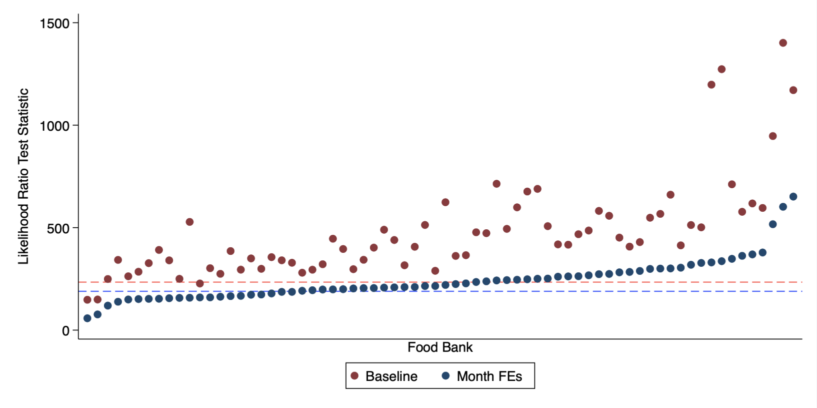


Note: Plot shows average bids across food banks × food types, controlling for available lots and endogenous entry. Estimates are sorted by average bid on dried food. 95% confidence intervals are shown.

return

Heterogeneity Over Time

Estimate food bank × type × month specific means, and consider LR statistic ◂ return



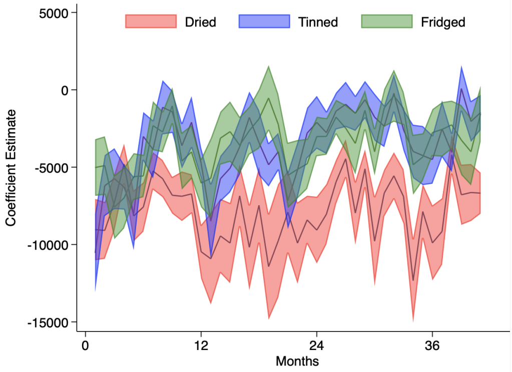
Note: This figure plots likelihood ratio test statistics for the hypothesis test that average bids for each type of food are constant over time, against the alternate hypothesis that bids vary by month. The estimated model controls for censoring, distance, and lot composition. The blue results also include month fixed effects. Under this null hypothesis the test statistic takes a χ^2 distribution with 200 (red) or 160 (blue) degrees of freedom. Critical values for tests at the 5% significance levels are plotted as horizontal lines.

Heterogeneity Over Time

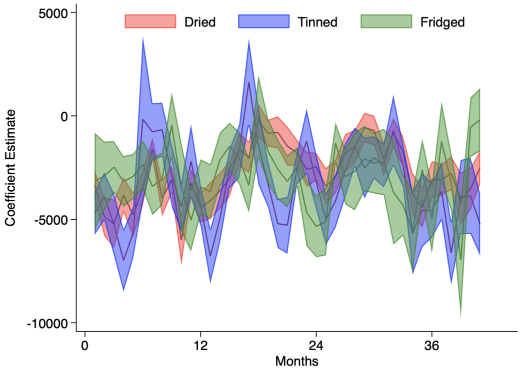
Instead, estimate α_{igm} = food bank $\times$ type $\times$ month specific means

◂ return

Food bank (A)



Food bank (B)



Note: Plots show average bids across food types $\times$ months, controlling for available lots and endogenous entry. The two food banks shown are the two highest consumption food banks ($\approx$ 5% of total food each). Fresh / Non-food are excluded for graph-ability. The shaded area gives the 95% confidence intervals.

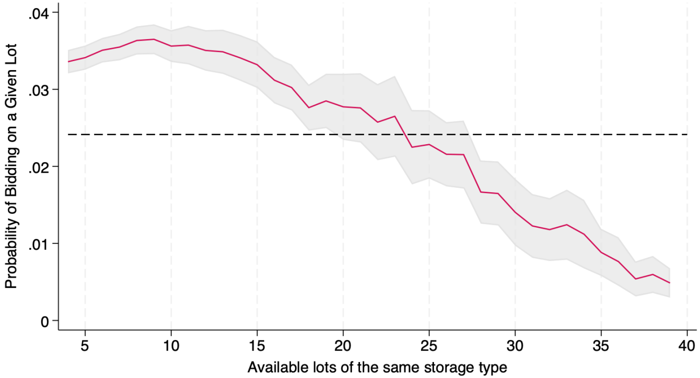
A Running Theme

A central theme will be this idea of heterogeneity:

1. Heterogeneity in food Graphs
   - Is Cereal qualitatively different from Frozen Dinners?
   - If food is all the same they will not care what they consume
2. Heterogeneity in needs across food banks Graphs
   - 5 food banks receive as much food as the 122 food banks that receive the least food
   - But, these 122 food banks spend 4 times as many shares
3. Heterogeneity in needs over time Graphs
   - Bidding behaviour within a food bank varies significantly over time

return

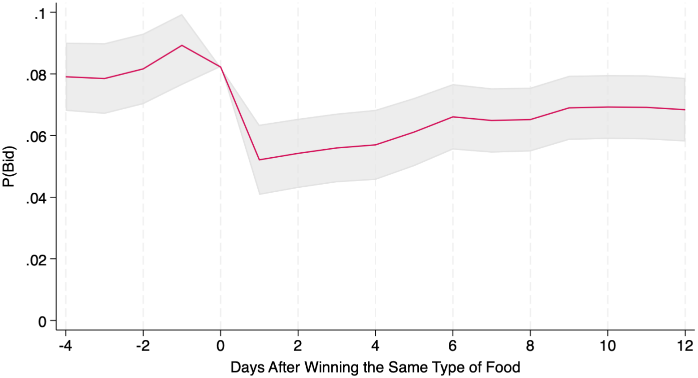
Static Substitution



Probability of bidding on a given lot, given available lots of the same storage type (±2 moving average). The dotted line gives the unconditional probability of bidding. Controlling for foodbank x storage method fixed effects, and day of week.

return

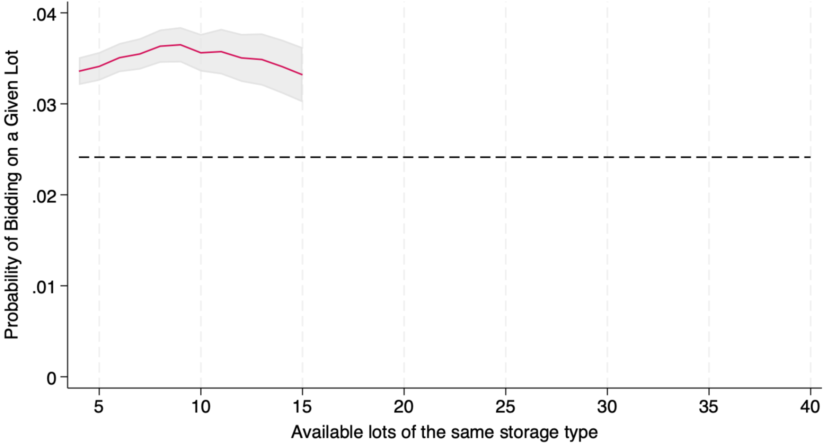
Dynamic Substitution



Controlling for foodbank x storage method fixed effects, conditional on bidding at t=0 and goods being available.
P(Bid) at day 0 is normalised to the long-run average probability of bidding, to demonstrate scale. t±1 moving average.

return

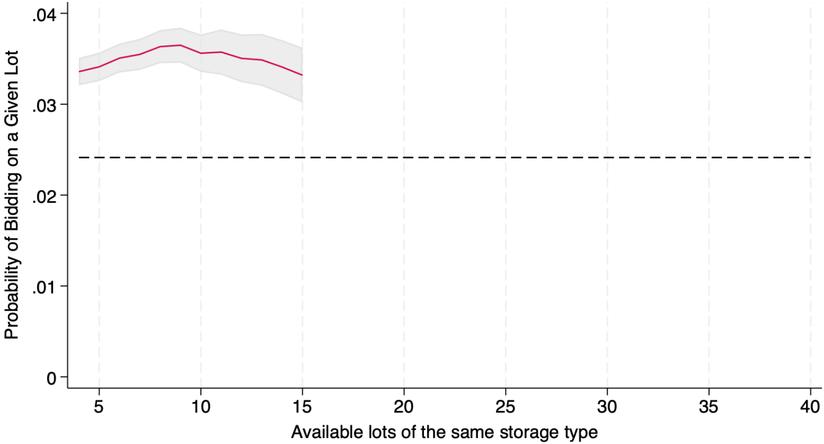
Identification of π_i



Probability of bidding on a given lot, given available lots of the same storage type (±2 moving average). The dotted line gives the unconditional probability of bidding. Controlling for foodbank x storage method fixed effects, and day of week.

◂ Identification Budget Constraint?

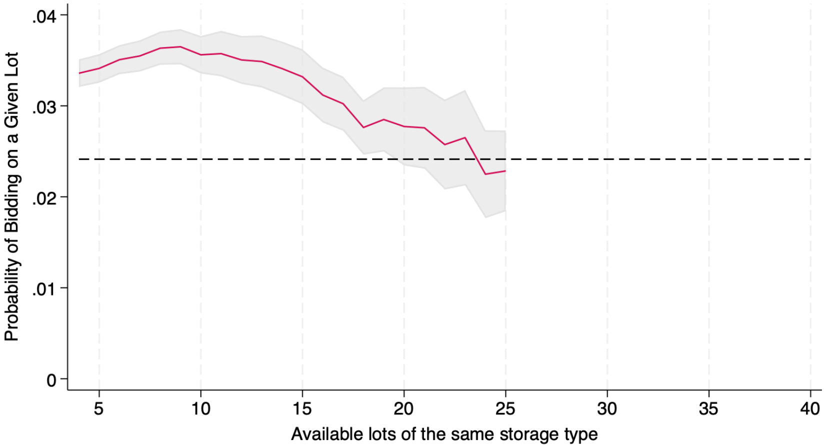
Identification of π_i



Probability of bidding on a given lot, given available lots of the same storage type (±2 moving average). The dotted line gives the unconditional probability of bidding. Controlling for foodbank x storage method fixed effects, and day of week.

◂ Identification Budget Constraint?

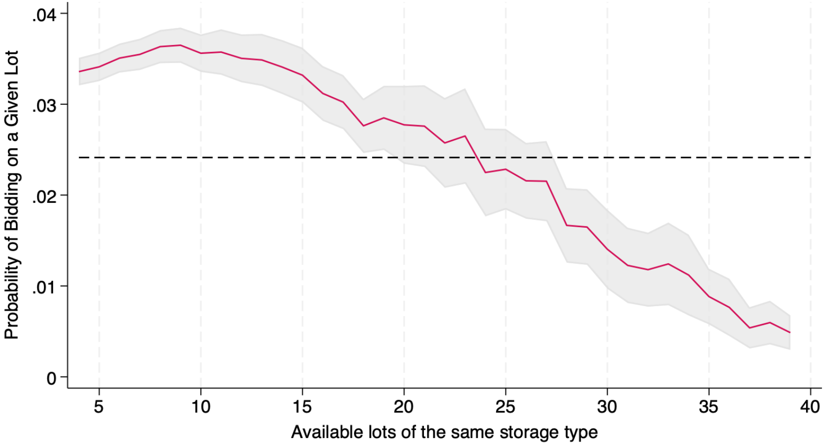
Identification of π_i



Probability of bidding on a given lot, given available lots of the same storage type (±2 moving average). The dotted line gives the unconditional probability of bidding. Controlling for foodbank x storage method fixed effects, and day of week.

◂ Identification   Budget Constraint?

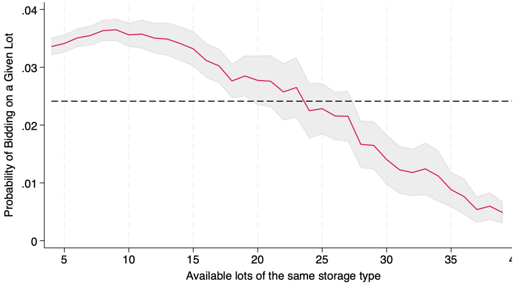
Identification of π_i



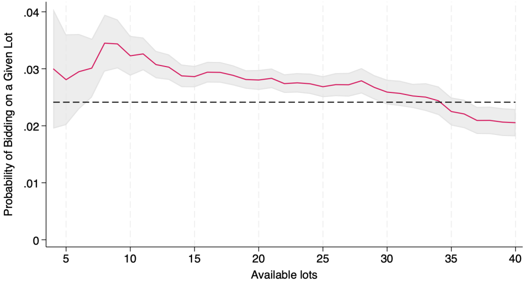
Probability of bidding on a given lot, given available lots of the same storage type (±2 moving average). The dotted line gives the unconditional probability of bidding. Controlling for foodbank x storage method fixed effects, and day of week.

◂ Identification · Budget Constraint?

Static Complementarities



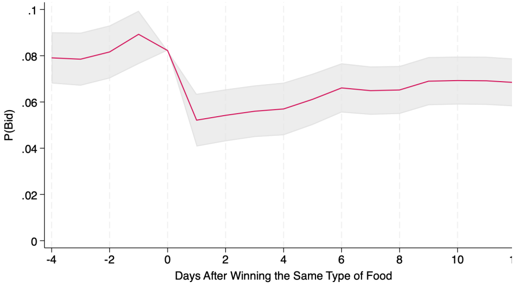
Probability of bidding on a given lot, given available lots of the same storage type (±2 moving average). The dotted li
gives the unconditional probability of bidding. Controlling for foodbank x storage method fixed effects, and day of wee



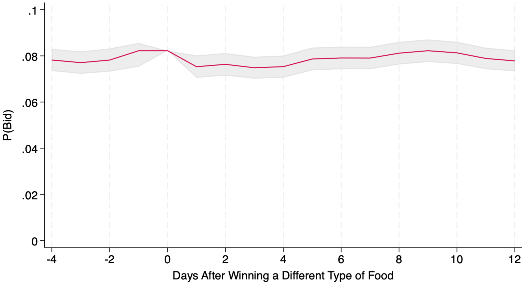
Probability of bidding on a given lot, given the total available lots (±4 moving average). The dotted line gives the unconditional probability of bidding. Controlling for foodbank x good fixed effects, and day of week.

return

Dynamic Complementarities



Controlling for foodbank x storage method fixed effects, conditional on bidding at t=0 and goods being available.
P(Bid) at day 0 is normalised to the long-run average probability of bidding, to demonstrate scale. t±1 moving averag

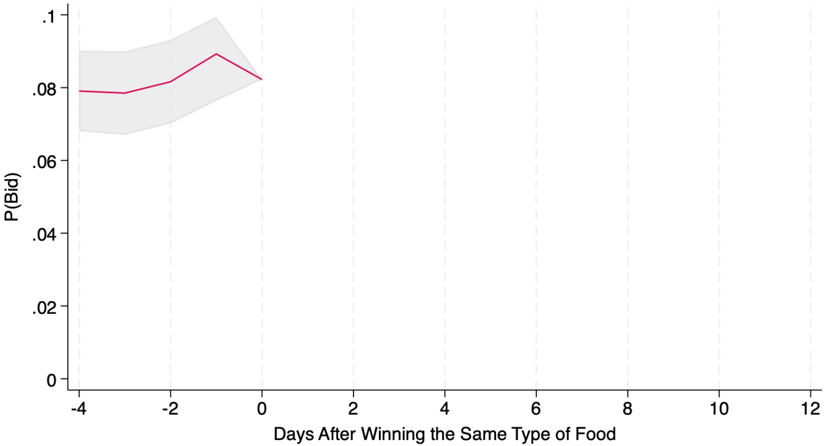


Controlling for foodbank x storage method fixed effects, conditional on goods being available. t±1 moving average.
P(Bid) at day 0 is normalised to the long-run average probability of bidding, to demonstrate scale.

Identification

Identification of F_i

$F_i : \mathbf{x}_{it} \sim N(\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}, \Sigma_i)$

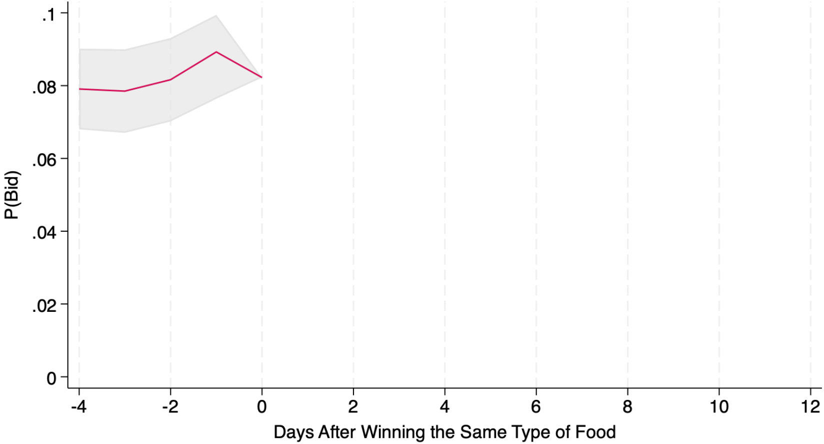


Controlling for foodbank x storage method fixed effects, conditional on bidding at t=0 and goods being available.
P(Bid) at day 0 is normalised to the long-run average probability of bidding, to demonstrate scale. t±1 moving average.

◀ Identification | Budget Constraint? | Big Food banks?

Identification of F_i

$F_i : \mathbf{x}_{it} \sim N(\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}, \Sigma_i)$

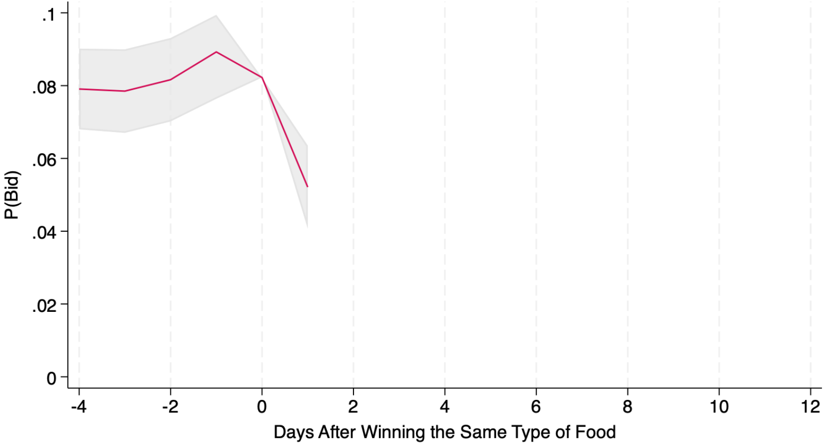


Controlling for foodbank x storage method fixed effects, conditional on bidding at t=0 and goods being available.
P(Bid) at day 0 is normalised to the long-run average probability of bidding, to demonstrate scale. t±1 moving average.

◂ Identification | Budget Constraint? | Big Food banks?

Identification of F_i

$$F_i : \mathbf{x}_{it} \sim N(\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}, \Sigma_i)$$

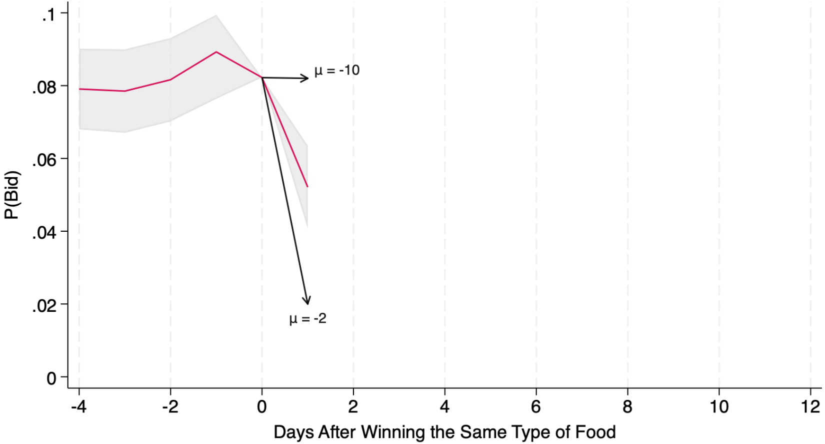


Controlling for foodbank x storage method fixed effects, conditional on bidding at t=0 and goods being available.
P(Bid) at day 0 is normalised to the long-run average probability of bidding, to demonstrate scale. t±1 moving average.

◀ Identification | Budget Constraint? | Big Food banks?

Identification of F_i

$$F_i : \mathbf{x}_{it} \sim N(\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}, \Sigma_i)$$

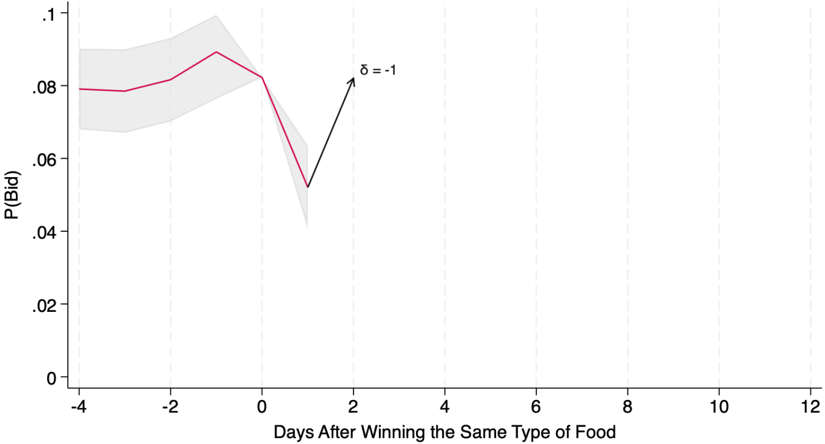


Controlling for foodbank x storage method fixed effects, conditional on bidding at t=0 and goods being available.
P(Bid) at day 0 is normalised to the long-run average probability of bidding, to demonstrate scale. t±1 moving average.

◂ Identification | Budget Constraint? | Big Food banks?

Identification of F_i

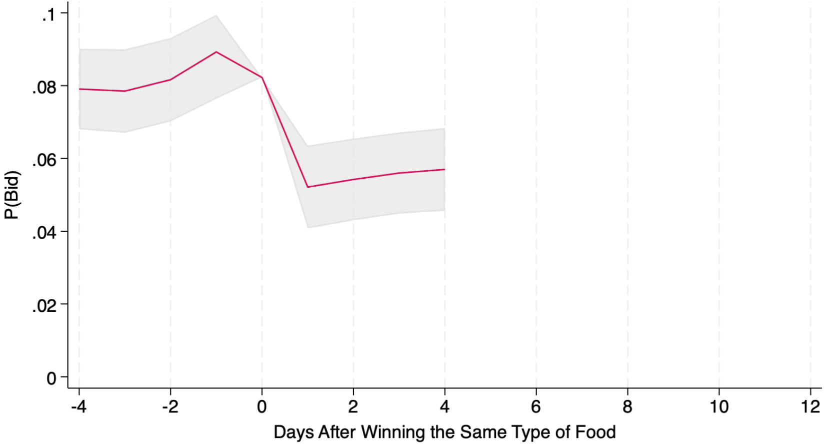
$$F_i : \mathbf{x}_{it} \sim N(\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}, \Sigma_i)$$



Controlling for foodbank x storage method fixed effects, conditional on bidding at t=0 and goods being available.
P(Bid) at day 0 is normalised to the long-run average probability of bidding, to demonstrate scale. t±1 moving average.

Identification of F_i

$F_i : \mathbf{x}_{it} \sim N(\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}, \Sigma_i)$

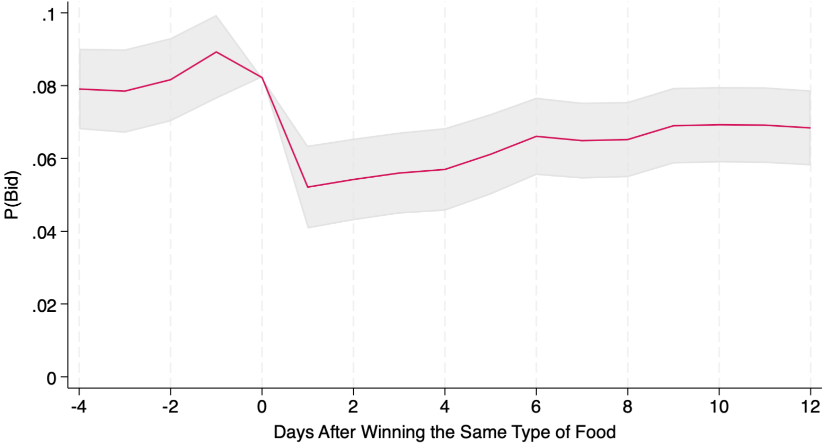


Controlling for foodbank x storage method fixed effects, conditional on bidding at t=0 and goods being available.
P(Bid) at day 0 is normalised to the long-run average probability of bidding, to demonstrate scale. t±1 moving average.

◀ Identification Budget Constraint? Big Food banks?

Identification of F_i

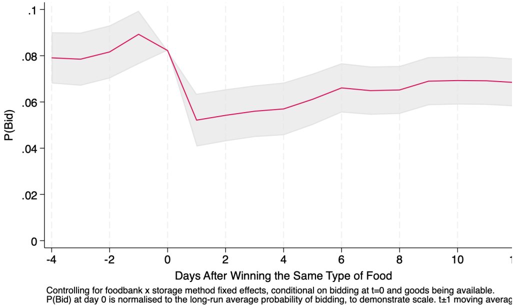
$F_i : \mathbf{x}_{it} \sim N(\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}, \Sigma_i)$

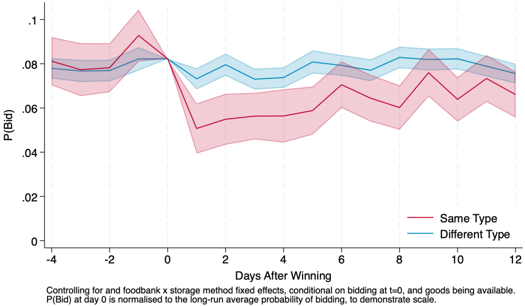


Controlling for foodbank x storage method fixed effects, conditional on bidding at t=0 and goods being available.
P(Bid) at day 0 is normalised to the long-run average probability of bidding, to demonstrate scale. t±1 moving average.

◂ Identification | Budget Constraint? | Big Food banks?

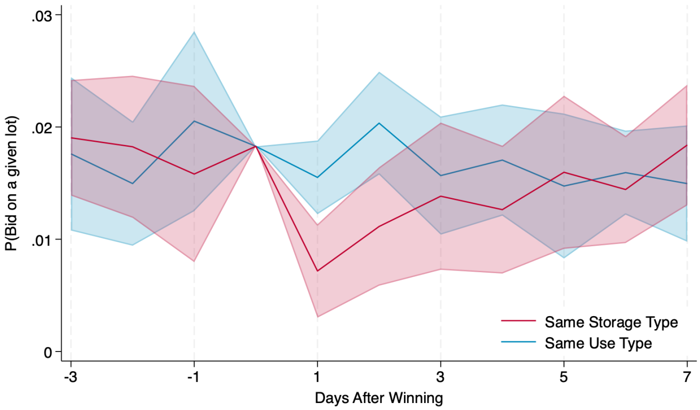
Dynamic Complementarities





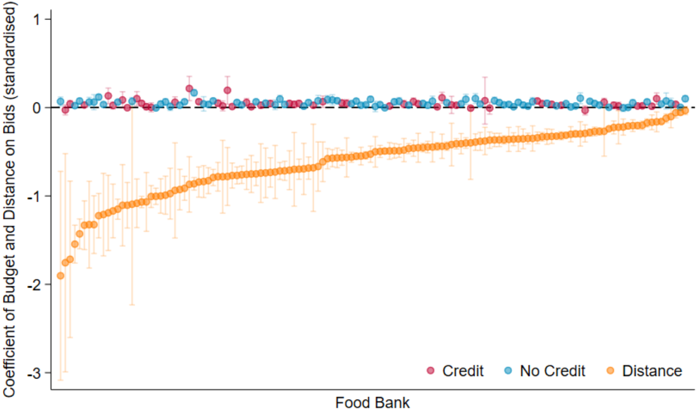
Identification

The Importance of Storage Costs



Note: plots the probability of bidding on a particular type of food, conditional on whether they won a load of the same storage type (in red) and conditional on whether they won a load of the same use type (in blue) at time zero.

Budgets do not Predict Bidding Behaviour



Note: plots coefficients of budget against bids across food banks, compared to the coefficients of distance against bids (orange). Excludes estimates that did not converge (all with credit access). Variables are standardised, includes food bank × month fixed effects.

◂ Model

Observational Equivalence of Quasi-Linearity and Budget Constraints

This proposition holds if:

- Intertemporal Budget Constraint: $E_t[\sum_{s=0}^{\infty} \frac{\sum_l b_{ilt+s} - y_{it+s}}{(1+r)^s}] = A_{it}$
- Continuity of expected payoffs in temporally dependent state variables $(\mathbf{s}_t, \mathbf{A}_t) = X_t$
- Weak Dependence: $\lim_{\tau\to\infty} F_{X_{t+\tau}|X_t} = F_{X_{t+\tau}}$

Proof.

- Lagrangian: $L_i(\mathbf{b}_t; X_t) =$
  $\sum_l \Gamma_l(b_{lt})\upsilon_{lt} + \sum_a P_a(\mathbf{b}_t)\kappa(X_t^a) - \lambda_t[\sum_l \Gamma_l(b_{lt}) - y_t - A_t + E_t \sum_{s=1}^{\infty} \frac{\sum_l b_{ilt+s} - y_{it+s}}{(1+r)^s}]$
- FOCs: $\lambda_t^*[\frac{\partial \Gamma_l}{\partial b_l} b_{lt}^* + \Gamma_l(b_{lt}^*)] = \frac{\partial \Gamma_l}{\partial b_l}\upsilon_l + \sum_a P_a(\mathbf{b}_t^*)\kappa(X_t^a)$, yields optimised $\lambda_t^* = \lambda(X_t)$
- Continuity of expected payoffs implies $\lambda(X_t)$ is continuous
- At the Optimum, $\lambda_t^* = E_t[\lambda_{t+s}^*|X_t]$ for $s > 0$. From Hall (1978). Otherwise...
- As we take $s \to \infty$ and impose weak dependence and continuity we get
  $\lambda_t^* = \int \lambda(X_{t+s})dF(X_{t+s}|X_t) = \int \lambda(X_{t+s})dF(X_{t+s}) = \lambda^*$ is constant over time
- Return to the FOCs with constant λ^*, these are the same as the Quasi-linear case.

$\square$

Model

Standard Approaches

The institutional setting is a challenge for **standard approaches**:

1. *Demand System Estimation* to find Compensating Variation
    - Accounting Period - Aggregate demand by month, or week?
    - Price Variation - Lack of within good type price variation
    - High Dimensional - A problem for Discrete Demand Estimation
    - Bid versus Win - Losing bids are not irrelevant

2. *Welfare Index Numbers* to find the Compensating Variation
    - Negative Prices/Satiation - Incompatible with most indices
    - Heterogeneity over time - Incompatible with most methods

3. *Sufficient Statistics* approach to estimating CV
    - Complexity - Only excessively simplistic models are tractable
    - Misses key variation - Difficult to introduce changes over time

return

Exogeneity of $\mathbf{x}$

- Essentially, I assume stocks *just happen* - $\mathbf{x}_{it}$ are an exogenous process
  then food banks respond by trying to win food on the Choice System

- This assumption likely biases my results **against** the value of choice
  - If there is reverse causality, can use winning to influence future net donations
  - Hence, additional benefits of allowing choice - more influence over net donations!
  - However, the effects on equity are more ambiguous → could be interesting to explore

- To an extent should be able to test this assumption using estimated donations

  → Look for correlation in net donations over time
  → Testing whether winnings Granger causes future net donations.

- I am also investigating whether I can do this as a robustness exercise
  - allow for reverse causation, or autocorrelation in net donations
  - This is possible in practice, but it is unclear whether any such process is identified

return

The Food Bank Model

- The set of food on offer
  - Pounds by storage method, $\mathbf{z}_{lt}^{g}$ and by Subcategory, $\mathbf{z}_{lt}^{h}$
  - $\mathbf{c}_{lt}$ - Other lot characteristics, e.g. location.

- States
  - Stock of each storage type $\mathbf{s}_{i}^{g}$
    $\rightarrow$ This helps me capture storage costs
  - Stock of each subcategory $\mathbf{s}_{i}^{h}$
    $\rightarrow$ But, assume $\pi(\mathbf{s}_i)$ is linear in $\mathbf{s}_{i}^{h}$, i.e. Constant Returns
    $\rightarrow$ Therefore the level of $\mathbf{s}_{i}^{h}$ doesn't matter, so I focus on changes through $\mathbf{z}_{lt}^{h}$
  - Aggregate supply $\mathbf{s}_0$: daily and previous 30 day supply, by storage type
    $\rightarrow$ This *might* impact $P(i \text{ wins } l|\mathbf{b})$

- Transition Function: $\mathbf{s}_{it}^{g} = \mathbf{s}_{it-1}^{g} + \mathbf{w}_{it-1}^{T}\mathbf{z}_{t-1}^{g} + \mathbf{x}_{it}$
  - This is **not** a random walk. It is closer to an error correction process
    $\rightarrow$ winnings $\mathbf{w}_{it-1}^{T}\mathbf{z}_{t-1}^{g}$ vary with $\mathbf{s}_{it-1}^{g}$ to prevent stocks dropping too low
  - I assume the process is stationary
    $\rightarrow$ This is an assumption about the competitive equilibrium
    $\rightarrow$ If $\mathbf{s}_{it}^{g}$ ends up as an $AR(1)$ process, I can actually test stationarity

return

Primitives (1)

return

- **Setup**
  - *Rules:* Player i wins lot l in period t if $b_{itl} \geq \max_{j \neq i} b_{jtl}$
  - *Reservation Prices* R_{tl} on each lot
  - *Entry* is costless - valuations are known before entering
  - *Ties* occur with zero probability*

Example: Two lots $\{apples, \ carrots\}$

Primitives (1)

return

- **Setup**
  - *Rules:* Player i wins lot l in period t if $b_{itl} \geq \max_{j \neq i} b_{jtl}$
  - *Reservation Prices* R_{tl} on each lot
  - *Entry* is costless - valuations are known before entering
  - *Ties* occur with zero probability*
- **States:**
  - Player i begins period t in state $\mathbf{s}_{it}$
  - Player i ends period t in state $\mathbf{s}_{it}^{a}$
  - superscript a refers to which combination of lots they ended up winning

Example:

- $\mathbf{s}_{it} = (stock_{it}^{apples}, stock_{it}^{carrots})$
- $\mathbf{s}_{it}^{a} = (stock_{it}^{apples} + winnings_{it}^{apples}, stock_{it}^{carrots} + winnings_{it}^{carrots})$

- **Setup**
  - *Rules:* Player i wins lot l in period t if $b_{itl} \geq \max_{j \neq i} b_{jtl}$
  - *Reservation Prices* R_{tl} on each lot
  - *Entry* is costless - valuations are known before entering
  - *Ties* occur with zero probability*

- **States:**
  - Player i begins period t in state $\mathbf{s}_{it}$
  - Player i ends period t in state $\mathbf{s}_{it}^{a}$
  - superscript a refers to which combination of lots they ended up winning

- **Lots**
  - $\mathbb{L}_t$ gives the set of lots players may bid on, with $max|\mathbb{L}| = L$
  - Lot l is described by a vector of characteristics $\mathbf{c}_{tl}$.

Example:
  - $\mathbb{L}_t \in \{\{\emptyset\}, \{apples\}, \{carrots\}, \{apples, \; carrots\}\}$
  - $\mathbf{c}_{t,apples}$ = (type of apples, location of apples)

Primitives (1)

return

- **Setup**
  - *Rules:* Player i wins lot l in period t if $b_{itl} \geq \max_{j \neq i} b_{jtl}$
  - *Reservation Prices* R_{tl} on each lot
  - *Entry* is costless - valuations are known before entering
  - *Ties* occur with zero probability*

- **States:**
  - Player i begins period t in state $\mathbf{s}_{it}$
  - Player i ends period t in state $\mathbf{s}^{a}_{it}$
  - superscript a refers to which combination of lots they ended up winning

- **Lots**
  - $\mathbb{L}_t$ gives the set of lots players may bid on, with $max|\mathbb{L}| = L$
  - Lot l is described by a vector of characteristics $\mathbf{c}_{tl}$.

- Denote the *Overall* state $\mathbf{s}_t = (\{\mathbf{s}_{it}\}_{i \in \mathbb{N}}, \mathbb{L}_t, \mathbf{C}_t)$

- **Valuations:**
  - *Lot specific:* $\boldsymbol{\upsilon}_{it} \sim F(.|\mathbf{s}_t)$, an L dimensional vector
  - *Combination Value:* $\pi_i(\mathbf{s}_{it})$, a 2^L dimensional vector
    Element a corresponds to ending period t in state $\mathbf{s}^a_{it}$: $\pi_i(\mathbf{s}^a_{it})$

Example:

- $\boldsymbol{\upsilon}_{it} = \begin{pmatrix} \upsilon_{it\ apples} \\ \upsilon_{it\ carrots} \end{pmatrix}$
- Or, if carrots are not available at t: $\boldsymbol{\upsilon}_{it} = \begin{pmatrix} \upsilon_{it\ apples} \\ . \end{pmatrix}$
- $\pi_i = \begin{pmatrix} J_{win\ nothing} \\ J_{win\ apples} \\ J_{win\ carrots} \\ J_{win\ apples\ \&\ carrots} \end{pmatrix}$

Where $J_{win\ apples\ \&\ carrots} \neq J_{win\ apples} + J_{win\ carrots}$

Primitives (2)

◂ return

- **Valuations:**
  - *Lot specific:* $\boldsymbol{v}_{it} \sim F(.|\mathbf{s}_t)$, an L dimensional vector
  - *Combination Value:* $\pi_i(\mathbf{s}_{it})$, a 2^L dimensional vector
    Element a corresponds to ending period t in state $\mathbf{s}^a_{it}$: $\pi_i(\mathbf{s}^a_{it})$

- **Actions:**
  - Player i chooses a subset of auctions to enter; $\mathbf{d}_{it}$
  - They then choose their bids conditional on entry; $\mathbf{b}_{it}$
  - *Strategies* conditional on primitives are given by σ_i

Example:

- If they only bid on apples: $\mathbf{d}_{it} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- If they only bid on apples: $\mathbf{b}_{it} = \begin{pmatrix} b_{it\ apples} \\ . \end{pmatrix}$

Primitives (2)

return

- **Valuations:**
  - *Lot specific:* $\boldsymbol{\upsilon}_{it} \sim F(.|\mathbf{s}_t)$, an L dimensional vector
  - *Combination Value:* $\pi_i(\mathbf{s}_{it})$, a 2^L dimensional vector
    Element a corresponds to ending period t in state $\mathbf{s}_{it}^a$: $\pi_i(\mathbf{s}_{it}^a)$
- **Actions:**
  - Player i chooses a subset of auctions to enter; $\mathbf{d}_{it}$
  - They then choose their bids conditional on entry; $\mathbf{b}_{it}$
  - *Strategies* conditional on primitives are given by σ_i
- **Equilibrium Win Probabilities:**
  - Player i wins lot l with probability $\Gamma_{il}(b_{itl}, d_{itl}; \sigma_{-i})$
  - $P_{ia}(\mathbf{b}_{it}, \mathbf{d}_{it}; \sigma_{-i})$ gives the probability of combination outcome a

Example:

$$P(\mathbf{b}_{it}, \mathbf{d}_{it}) = \begin{pmatrix} P(\text{win nothing}|\mathbf{b}_{it}, \mathbf{d}_{it}) \\ P(\text{win apples only}|\mathbf{b}_{it}, \mathbf{d}_{it}) \\ P(\text{win carrots only}|\mathbf{b}_{it}, \mathbf{d}_{it}) \\ P(\text{win both}|\mathbf{b}_{it}, \mathbf{d}_{it}) \end{pmatrix} = \begin{pmatrix} [1-\Gamma_{apples}(b_a)][1-\Gamma_{carrots}(b_c)] \\ \Gamma_{apples}(b_a)[1-\Gamma_{carrots}(b_c)] \\ [1-\Gamma_{apples}(b_a)]\Gamma_{carrots}(b_c) \\ \Gamma_{apples}(b_{apples})\Gamma_{carrots}(b_{carrots}) \end{pmatrix}$$

Primitives (2)

return

- **Valuations:**
  - *Lot specific:* $\boldsymbol{\upsilon}_{it} \sim F(.|\mathbf{s}_t)$, an L dimensional vector
  - *Combination Value:* $\pi_i(\mathbf{s}_{it})$, a 2^L dimensional vector
    Element a corresponds to ending period t in state $\mathbf{s}^a_{it}$: $\pi_i(\mathbf{s}^a_{it})$
- **Actions:**
  - Player i chooses a subset of auctions to enter; $\mathbf{d}_{it}$
  - They then choose their bids conditional on entry; $\mathbf{b}_{it}$
  - *Strategies* conditional on primitives are given by σ_i
- **Expected Payoffs:** $EU(\mathbf{b},\mathbf{d}|\boldsymbol{\upsilon}_i,\mathbf{s};\sigma_{-i}) =$

$$\Gamma_i(\mathbf{b},\mathbf{d};\sigma_{-i})^T(\boldsymbol{\upsilon}_i - \mathbf{b}) + P_i(\mathbf{b},\mathbf{d};\sigma_{-i})^T[\pi_i(\mathbf{s}) + \beta V_i(\mathbf{s};\sigma_{-i})]$$

Example: $EU(\mathbf{b},\mathbf{d}|\boldsymbol{\upsilon}_i,\mathbf{s};\sigma_{-i}) =$

$$\begin{pmatrix}\Gamma_a(b_a)\\ \Gamma_c(b_c)\end{pmatrix}^T \begin{pmatrix}\upsilon_{it\ apples} - b_a\\ \upsilon_{it\ carrots} - b_c\end{pmatrix} + \begin{pmatrix}P(\text{nothing}|\mathbf{b}_{it},\mathbf{d}_{it})\\ P(\text{apples}|\mathbf{b}_{it},\mathbf{d}_{it})\\ P(\text{carrots}|\mathbf{b}_{it},\mathbf{d}_{it})\\ P(\text{both}|\mathbf{b}_{it},\mathbf{d}_{it})\end{pmatrix}^T \begin{pmatrix}J_{win\ nothing} + \beta V_{win\ nothing}\\ J_{win\ apples} + \beta V_{win\ apples}\\ J_{win\ carrots} + \beta V_{win\ carrots}\\ J_{win\ both} + \beta V_{win\ both}\end{pmatrix}$$

Bellman equation (Mulit-Object, endogenous entry, reservation prices)

$$W(\boldsymbol{\upsilon}, \mathbf{s}; \sigma_{-i}) =$$

$$\max_{\mathbf{b},\mathbf{d}} \left\{ \begin{array}{l} \sum_{l \in \mathbb{L}(\mathbf{s})} \Gamma_l(\mathbf{b}, \mathbf{d}|\mathbf{s})(\upsilon_l - b_l) + \sum_{\mathbf{w}^a \in \mathbb{W}(\mathbf{s})} P_a(\mathbf{b}, \mathbf{d}|\mathbf{s})\pi(\mathbf{s}^a) \\ \\ \quad + \beta \sum_{\mathbf{w}^a} P_a(\mathbf{b}, \mathbf{d}|\mathbf{s}) \int_{\tilde{\mathbf{s}}} \int_{\tilde{\boldsymbol{\upsilon}}} W(\tilde{\boldsymbol{\upsilon}}, \tilde{\mathbf{s}}); \sigma_{-i}) dF(\tilde{\boldsymbol{\upsilon}}|\tilde{\mathbf{s}}) dT(\tilde{\mathbf{s}}|\mathbf{s}^a) \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad s.t. \quad b_l \geq R_l \ \ \forall l \end{array} \right\}$$

Where:

- $\Gamma_l(\mathbf{b}, \mathbf{d}|\mathbf{s}) = P(\text{win lot } l|\mathbf{b}, \mathbf{d}; \mathbf{s})$
- $P_a(\mathbf{b}, \mathbf{d}|\mathbf{s}) = P(\text{the combination outcome is } a|\mathbf{b}, \mathbf{d}; \mathbf{s})$
- $\mathbf{s}^a$ gives the ex-post state from combination outcome a

return

Equilibrium

- Focus on *Markov Perfect Equilibria* in Symmetric Markovian Strategies
  1. Agent's maximise the NPV of payoffs, given beliefs
  2. Beliefs are consistent with observed play

  $\rightarrow \sigma_i$ depends only on $\mathbf{s}_t$, not on t itself

- *Equilibrium Existence*
  - Conditional on the existence of a Bayesian Nash Equilibrium in the Stage Game, taking entry as given, a Markov Perfect Equilibrium exists.

    (For the quasi-linear utility case)
    For the proof, see my first chapter
  - However, equilibrium in the Stage Game has yet to be proven

    Except for specific cases of preferences
  - But this is not a practical problem
    - In practice, I assume food banks have beliefs consistent with observed behaviour
    - This is consistent with a number of non-standard equilibrium models
      e.g. Backus & Lewis (2016)

return

Proposition (*Non-Parametric Point Identification*)

Assuming that:

1. *For any set of available lots* $\mathbf{s}_{0t}$ *we have that:* $E[g(\mathbf{s}_t)|\mathbf{s}_{0t}, \mathbf{b}_t] = 0$ *implies* $g(\mathbf{s}_t) = 0$ *for any real bounded function* g.
2. *For any* $\mathbf{s}_{0t+1}$ *there exists* $(\mathbf{w}_t, \mathbf{s}_{0t})$ *and neighbourhood* $(\bar{\mathbb{W}}, \bar{\mathbb{S}}_0)$ *around* $(\mathbf{w}_t, \mathbf{s}_{0t})$ *s.t. for* $(\bar{\mathbf{w}}_t, \bar{\mathbf{s}}_{0t}) \in (\bar{\mathbb{W}}, \bar{\mathbb{S}}_0)$*:* $E[g(\mathbf{b}_{t+1})|\mathbf{s}_{0t+1}, \bar{\mathbf{w}}_t, \bar{\mathbf{s}}_{0t}, \mathbf{w}_{t-1}] = 0$ *implies* $g(\mathbf{b}_{t+1}) = 0$.

Then $\pi_i(\mathbf{s})$, F_i^x, *and* F_i^υ *are point identified*

Proposition (*Non-Parametric Point Identification*)

Assuming that:

1. *For any set of available lots* $\mathbf{s}_{0t}$ *we have that:* $E[g(\mathbf{s}_t)|\mathbf{s}_{0t}, \mathbf{b}_t] = 0$ *implies* $g(\mathbf{s}_t) = 0$ *for any real bounded function* g*.*
2. *For any* $\mathbf{s}_{0t+1}$ *there exists* $(\mathbf{w}_t, \mathbf{s}_{0t})$ *and neighbourhood* $(\bar{\mathbb{W}}, \bar{\mathbb{S}}_0)$ *around* $(\mathbf{w}_t, \mathbf{s}_{0t})$ *s.t. for* $(\bar{\mathbf{w}}_t, \bar{\mathbf{s}}_{0t}) \in (\bar{\mathbb{W}}, \bar{\mathbb{S}}_0)$*:* $E[g(\mathbf{b}_{t+1})|\mathbf{s}_{0t+1}, \bar{\mathbf{w}}_t, \bar{\mathbf{s}}_{0t}, \mathbf{w}_{t-1}] = 0$ *implies* $g(\mathbf{b}_{t+1}) = 0$*.*

Then $\pi_i(\mathbf{s})$, F_i^x, *and* F_i^υ *are point identified*

Proof Outline:

1. Altmann (WP) shows that π and F^υ follow from $F_{\mathbf{b}_t|\mathbf{s}_t,\mathbf{s}_{0t}}$ and $F_{\mathbf{s}_t,\mathbf{s}_{0t}|\mathbf{s}_{t-1},\mathbf{s}_{0t-1}}$.
2. I then use machinery* from Hu and Shum (2012) to show that $F_{\mathbf{b}_t|\mathbf{s}_{0t},\mathbf{w}_{t-1},\mathbf{s}_{0t-1},\mathbf{s}_{t-1}}$ are identified using a Spectral Decomposition of $F_{\mathbf{b}_{t+1}),\mathbf{s}_{0t+1},\bar{\mathbf{w}}_t,\bar{\mathbf{s}}_{0t},\mathbf{w}_{t-1}}$.

   → I require existence of observed shifter $\mathbf{w}$, excluded from $F_{\mathbf{b}|\mathbf{s}_0,\mathbf{s}}$
   → But, I require weaker normalising assumptions on $\mathbf{s}$
3. $F_{\mathbf{b}_t|\mathbf{s}_t,\mathbf{s}_{0t}}$ and $F_{\mathbf{s}_t,\mathbf{s}_{0t}|\mathbf{s}_{t-1},\mathbf{s}_{0t-1}}$ are then separately identified from $F_{\mathbf{b}_t|\mathbf{s}_{0t},\mathbf{w}_{t-1},\mathbf{s}_{0t-1},\mathbf{s}_{t-1}}$ and $F_{\mathbf{b}_t,\mathbf{s}_{0t},\mathbf{s}_{0t-1},\mathbf{w}_{t-1}}$.

Identification

This estimation procedure is unusual, but used for tractability

- In the Auction literature: Assume a bid distribution then back out j & V
  - This is not possible on account of the unobserved state / Multi-object environment
  - We cannot write V as a distribution of bids **only**

- In the Dynamic Discrete Choice literature: Given functional form for j, solve for V
  - Assume a form for J, solve for V in each likelihood evaluation
  - Solving for V given j requires numerically finding optimal $\mathbf{b}^*$, which is slow

- CCP methods: Given observed actions (bids, entry) solve for V
  - We cannot write V as a distribution of bids **only**
  - But in a discrete choice context, fitting a parametric functional form to CCPs is numerically equivalent to this 3 step method.

return

Estimation Step 1

Estimate $P(i \text{ wins } l|\mathbf{b}) = \Gamma_l(b_{il})$:

- For lot l with characteristics $\mathbf{c}_{lt}$

$$\Gamma_{il}(x|\mathbf{c}_{lt}, \mathbf{s}_t) = GEV(x; \alpha_c, \sigma_c, \mu_h + \vartheta_g \mathbf{s}_0) \tag{2}$$

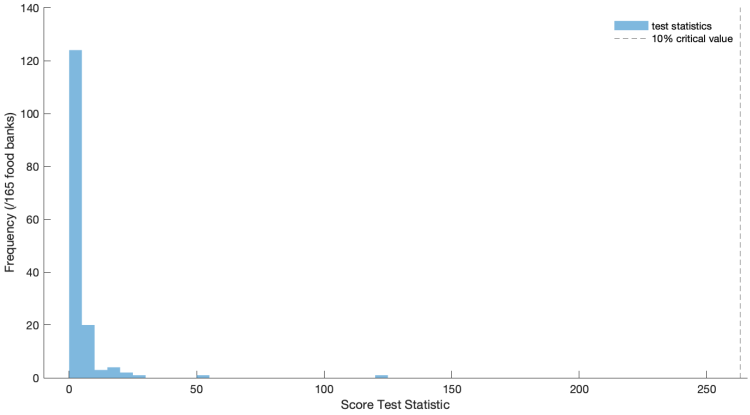
- Shape and Scale parameters ξ_c and ζ_c, category specific
- Subcategory specific mean parameter ν_{ch}
- Storage type specific log-linear 'demand' parameters ν_g
  - $\rightarrow$ Allowing the distribution of winning bids to vary with recent supply, by type of food
  - $\rightarrow$ Key Assumption: Γ_l is independent of $\mathbf{s}_i$. e.g. Large Market.
- The distribution is 'censored' at the reservation price

GEV assumption is motivated by extreme value theory

- $P(i \quad wins|b_i)$ is equivalent to b_i is the highest bid
- i.e. The extreme value

return

Large Market Assumption



Note: Plots the LR test statics, testing whether the distribution of winning bids changes as we drop data from individual food banks. This shows that variation in bidding behaviour by a single food bank cannot shift the distribution of winning bids. Therefore, variation in $\mathbf{s}_i$ cannot shift this distribution either.

return

Estimation Step 2 (a)

- Assume parametric form $k_i(\mathbf{s}_i) = \Phi \mathbf{s}_i^h + \mathbf{s}_i^{gT} \Psi_i \mathbf{s}_i^g$
  - Φ are assumed common across i - food banks have the same 'utility'
  - Ψ_i vary across i and negative definite, similar to quadratic storage costs.
    $\rightarrow$ More like 'opportunity cost' in this dynamic context
    $\rightarrow$ Different food banks likely have different storage costs, but very different opportunity costs

- I make the following distributional assumptions:
  - $\upsilon_{ilt} \sim N(\alpha_i \text{distance}_{ilt}, \sigma_c^2)$
  - $\mathbf{x}_{it} \sim N(\boldsymbol{\mu}_i, \Sigma_i)$

- The bidder's maximisation problem yields the optimality condition:

$$\lambda_i(b_{ilt} + \frac{\Gamma_l(b_{ilt})}{\nabla \Gamma_l(b_{ilt})}) \geq \upsilon_{ilt} + \Phi \mathbf{z}_{lt}^h + \mathbf{z}_{lt}^{gT} \Psi_i (\mathbf{z}_{lt}^g + 2\mathbf{s}_{it}^g + 2 \sum_{m \neq l} \Gamma_m(b_{imt}) \mathbf{z}_{mt}^g) \qquad (3)$$

Which holds with equality when $b_{ilt} > R_l$
$\rightarrow$ If I observed $\mathbf{s}_{it}^g$ this would just be a censored regression equation!

return

Estimation Step 2 (b)

I have a three equation (censored) Linear Gaussian State Space model:

$$\mathbf{s}^g_{it} = \mathbf{s}^g_{it-1} + \mathbf{w}^T_{it-1}\mathbf{z}^g_{t-1} + \mathbf{x}_{it} \qquad \rightarrow \text{ Transition Eq.}$$

$$\lambda_i y_{ilt} = \upsilon_{ilt} + \Phi\mathbf{z}^h_{lt} + \mathbf{z}^{gT}_{lt}\Psi_i(\mathbf{z}^g_{lt} + 2\mathbf{s}^g_{it} + 2\sum_{m\neq l}\Gamma_m(b_{imt})\mathbf{z}^g_{mt}) \qquad \rightarrow \text{ Observation Eq.}$$

$$y_{ilt} = \begin{cases} b_{ilt} + \frac{\Gamma_l(b_{ilt})}{\nabla\Gamma_l(b_{ilt})} & \text{if } b_{ilt} > R_l \\ R_l & \text{otherwise} \end{cases} \qquad \rightarrow \text{ Censoring Eq.} \quad (4)$$

Estimation is done using a Gibbs Sampler:

- We want to draw samples of (θ_1, θ_2) from its posterior; $f(\theta_1, \theta_2 | data)$
  $\rightarrow$ but sampling from this distribution is hard.
- Instead, we can iteratively draw samples from $f(\theta_1|\theta_2, data)$ and $f(\theta_2|\theta_1, data)$
  $\rightarrow$ these conditional samples approximate the posterior distribution

return

Estimation Step 2 (b)

I have a three equation (censored) Linear Gaussian State Space model:

$$\mathbf{s}_{it}^{g} = \mathbf{s}_{it-1}^{g} + \mathbf{w}_{it-1}^{T}\mathbf{z}_{t-1}^{g} + \mathbf{x}_{it} \qquad \rightarrow \text{ Transition Eq.}$$

$$\lambda_i y_{ilt} = \upsilon_{ilt} + \Phi\mathbf{z}_{lt}^{h} + \mathbf{z}_{lt}^{gT}\Psi_i(\mathbf{z}_{lt}^{g} + 2\mathbf{s}_{it}^{g} + 2\sum_{m\neq l}\Gamma_m(b_{imt})\mathbf{z}_{mt}^{g}) \qquad \rightarrow \text{ Observation Eq.}$$

$$y_{ilt} = \begin{cases} b_{ilt} + \frac{\Gamma_l(b_{ilt})}{\nabla\Gamma_l(b_{ilt})} & \text{if } b_{ilt} > R_l \\ R_l & \text{otherwise} \end{cases} \qquad \rightarrow \text{ Censoring Eq.} \quad (4)$$

Estimation is done using a Gibbs Sampler:

1. Given beliefs Γ, parameters of the pseudo-static model $\{k_i, F_i^{\upsilon}, F_i^{\mathbf{x}}\}$, and states $\{\mathbf{s}_{it}^{g}\}$:
   $\rightarrow$ draw censored values of $\{y_{ilt}\}$ using the Censoring Equation
2. Given Beliefs Γ, $\{k_i, F_i^{\upsilon}, F_i^{\mathbf{x}}\}$, and censored observations $\{y_{ilt}\}$:
   $\rightarrow$ draw $\{\mathbf{s}_{it}^{g}\}$ using the Transition / Observation equations (Carter-Kohn)
3. Given beliefs Γ, censored observations $\{y_{ilt}\}$, and states $\{\mathbf{s}_{it}^{g}\}$:
   $\rightarrow$ draw $\{k_i, F_i^{\upsilon}, F_i^{\mathbf{x}}\}$ from their posterior using the Observation Equation.
4. Repeat

return

Estimation Step 3

Write the continuation value $V(\mathbf{s})$ as a function of Γ, F^{υ}, F^{x} and k:

Proposition

The ex-ante Value Function can be expressed as:

$$E[W(\upsilon, \mathbf{s}_i^g)|\mathbf{s}_i^g] = \frac{E[q_t(\mathbf{s}_i^g)\pi(\mathbf{b}_{it}|\mathbf{s}_i^g)]}{E[q_t(\mathbf{s}_i^g)]}$$

Where $q_t(\mathbf{s}_i^g)$ *gives the posterior probability that* $\mathbf{s}_{it}^g = \mathbf{s}_i^g$ *and*

$$\pi(\mathbf{b}|\mathbf{s}_i^g) = \sum_l \lambda_i \frac{\Gamma_l(b_{il})^2}{\nabla_b \Gamma_l(b_{il})} - \sum_{m \neq l} \Gamma_l(b_{il}) \mathbf{z}_l^{g\,T} \Psi_i \mathbf{z}_m^g \Gamma_m(b_{im}) + \mathbf{s}_i^{g\,T} \Psi_i \mathbf{s}_i^g$$

$\rightarrow$ This is essentially just an extension of Arcidiacono & Miller (2011)

- The continuation value is given by $V(\mathbf{s}_i^g) = \int E[W(\upsilon, \mathbf{s}_i^g + \mathbf{x})|\mathbf{s}_i^g + \mathbf{x}]dF^{x}(\mathbf{x})$ and, finally, $\pi(\mathbf{s}_i^g) = k(\mathbf{s}_i^g) - \beta V(\mathbf{s}_i^g)$
- Evaluate these expressions for a sample of parameters, drawn from their posterior

return

The Estimation Procedure in a Nutshell

- The standard procedure fails because of the multi-object environment:
  $\rightarrow$ We cannot write the Value Function as a function of bids only

The Estimation Procedure in a Nutshell

- Altmann (WP) demonstrates that we *can* write the value function as a function of:
  1. Beliefs about Γ and P, the equilibrium win probabilities
  2. The Pseudo-Primitives of a 'misspecified' static game: $k_i(\mathbf{s}) = \pi_i(\mathbf{s}_i) + \beta V(\mathbf{s})$

  $\rightarrow$ Therefore, if we 'knew' $k_i(\mathbf{s})$ (and F_i^x), we could evaluate $V(\mathbf{s})$, then back out $\pi_i(\mathbf{s}_i)$
- The main problem is how do we estimate $k_i(\mathbf{s})$ and F_i^x?

return results

The Estimation Procedure in a Nutshell

- Altmann (WP) demonstrates that we *can* write the value function as a function of:
  1. Beliefs about Γ and P, the equilibrium win probabilities
  2. The Pseudo-Primitives of a 'misspecified' static game: $k_i(\mathbf{s}) = \pi_i(\mathbf{s}_i) + \beta V(\mathbf{s})$

  $\rightarrow$ Therefore, if we 'knew' $k_i(\mathbf{s})$ (and F_i^x), we could evaluate $V(\mathbf{s})$, then back out $\pi_i(\mathbf{s}_i)$

- The main problem is how do we estimate $k_i(\mathbf{s})$ and F_i^x?

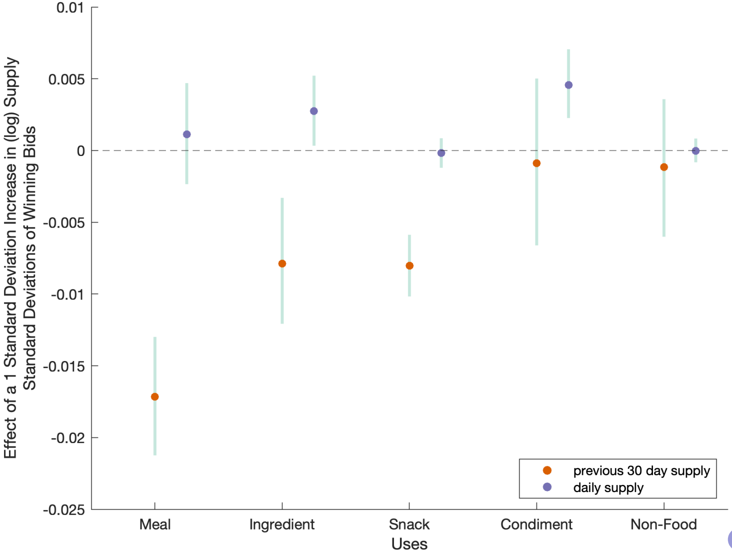
  $\rightarrow$ I make parametric assumptions on F_i^x and $k_i(\mathbf{s}_i)$:
  - $\mathbf{x}_{it}$ is normally distributed, with mean $\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}$ and variance Σ_i
  - $k_i(\mathbf{s}_i)$ is quadratic in $\mathbf{s}_i$ (Linear Demand Curves)

- The general idea behind estimation:
  1. Observe how behaviour changes with the available lots, or around a recent win
  2. Consider other changes in bidding behaviour...

     ...and find the change in stocks that rationalise this behaviour

return results

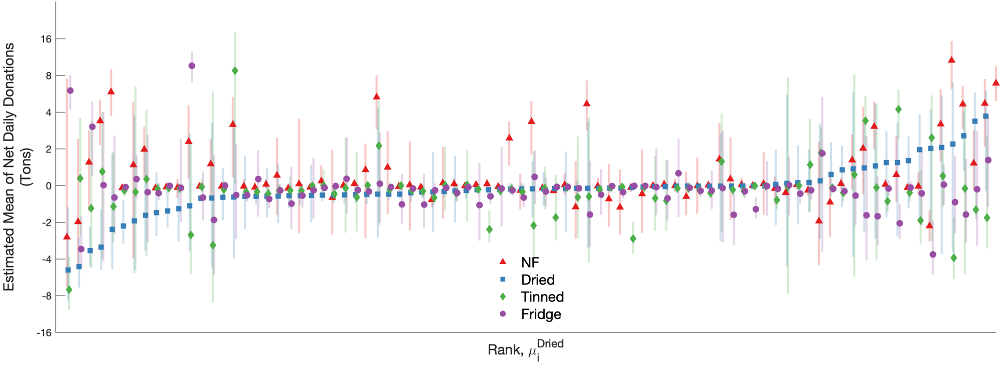
First Stage Demand results



return

Results: Net Donations

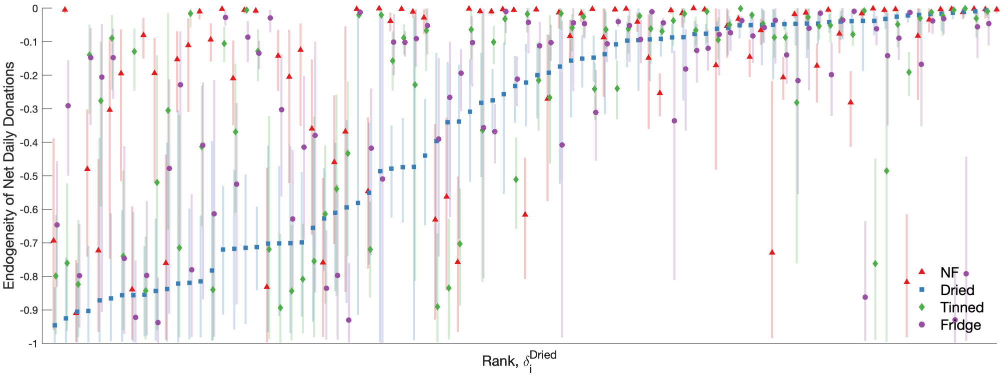
Estimated mean net donations ($\boldsymbol{\mu}_i = \hat{E}[\mathbf{x}_{it}]$)



Note: Plot shows estimated mean net local donations by food bank × food type, sorted across food banks by estimate for Dried food (blue). Error bars give 95% credible intervals. return

Estimated $\hat{\delta}$, the relationship between (lagged) stocks and net donations:

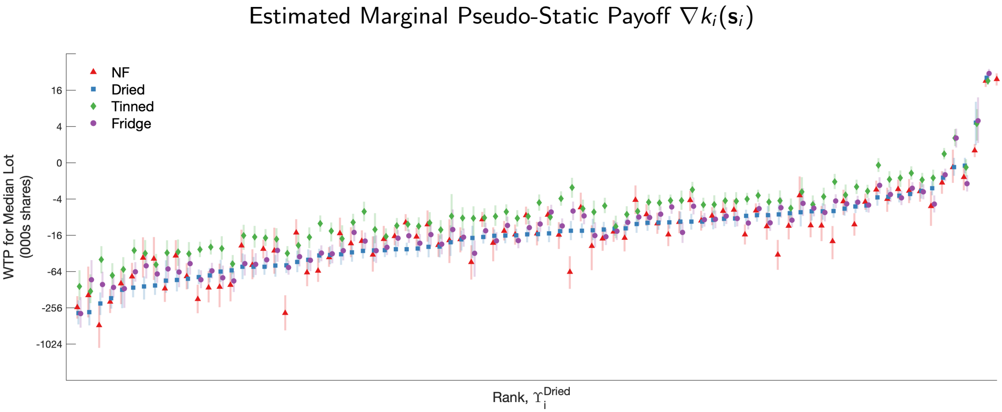
$$\mathbf{s}_{it} = \mathbf{s}_{it-1} + \mathbf{w}_{it-1} + \mathbf{x}_{it}, \text{ Where } \mathbf{x}_{it} \sim N(\boldsymbol{\mu}_i + \boldsymbol{\delta}_i \mathbf{s}_{it-1}, \Sigma_i)$$



Note: Plot shows estimated coefficient of lagged stocks on current net donations by food bank $\times$ food type, sorted across food banks by estimate for Dried food (blue). Error bars give 95% credible intervals. *Estimates* $+1$ can be interpreted as AR(1) coefficients, so that 0 implies strong persistence in stocks, and -1 implies no persistence.

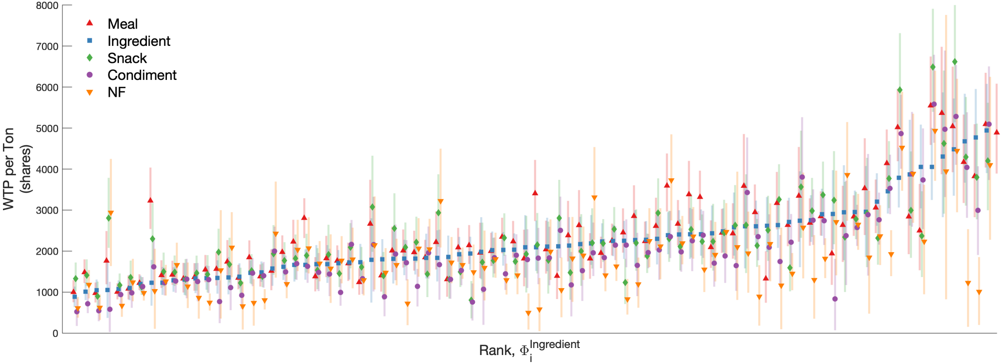
return

Results: Demand



Note: Plot shows estimated marginal pseudo-static pay-off from receiving an average lot by food bank $\times$ food type, evaluated when stocks are at their long run average. Estimates are sorted across food banks by estimate for Dried food. 95% credible intervals are plotted. return

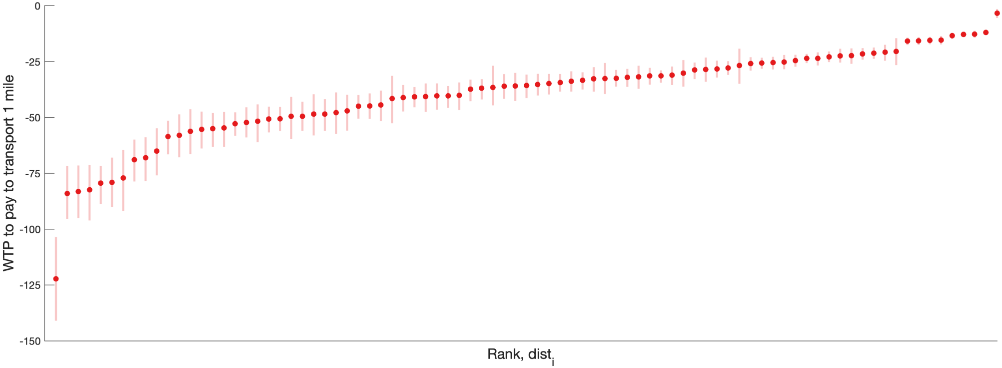
Estimated Marginal Pseudo-Static Payoff $\nabla k_i(\mathbf{s}_i)$, with respect to food usage.



Note: Plot shows estimated marginal pseudo-static pay-off from receiving an average lot by food bank $\times$ food type. This shows the benefit food banks get from having food in stock, to be given out to clients. Estimates are sorted across food banks by estimate for food that is an ingredient. 95% credible intervals are plotted.

return

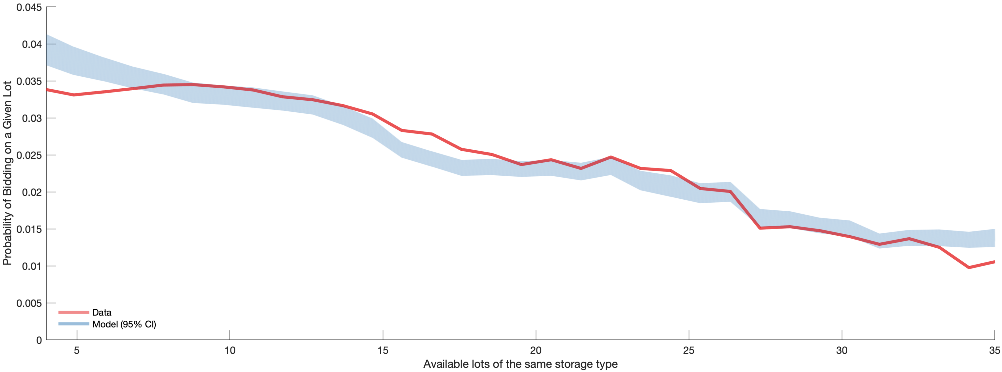
Results: Estimated Transport Costs



Note: Plot shows estimated transport costs: The number of additional shares a food bank is willing to pay for a lot that is one km further away. Estimates are sorted across food banks. 95% credible intervals are plotted.

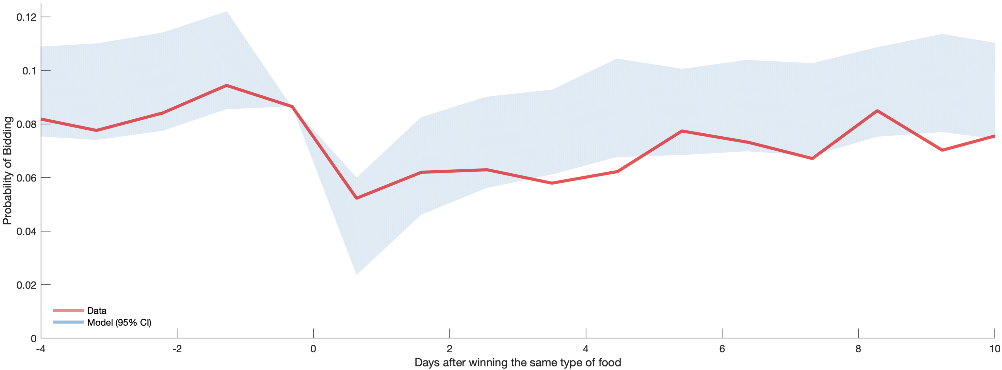
return

Results: Model Fit (1)



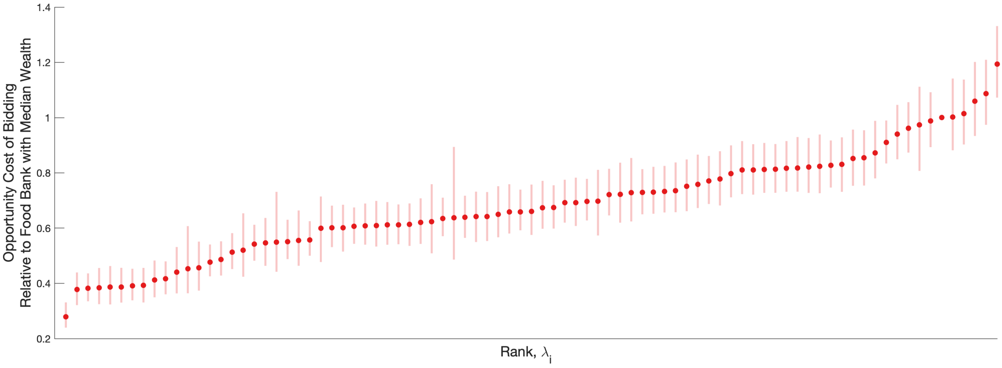
Note: The shaded area gives 95% credible intervals of the simulated data, while the red line gives mean estimates from the true data. Standard errors for the true data are not plotted. return

Results: Model Fit (2)



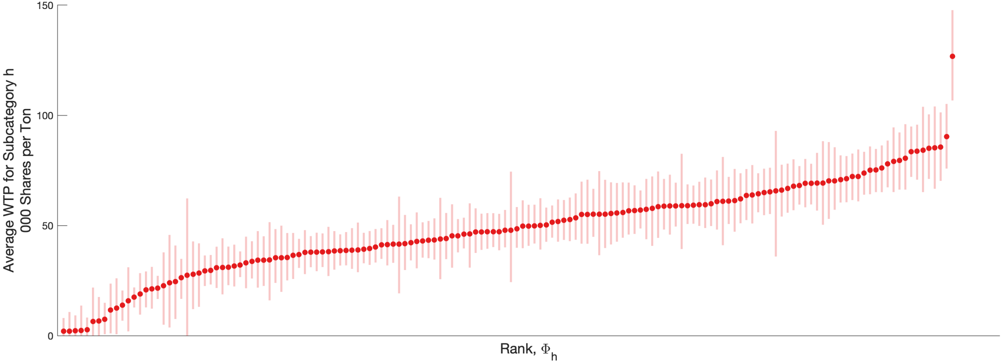
Note: The shaded area gives 95% credible intervals of the simulated data, while the red line gives mean estimates from the true data. Standard errors for the true data are not plotted. return

Results: Estimated Marginal Value of Wealth



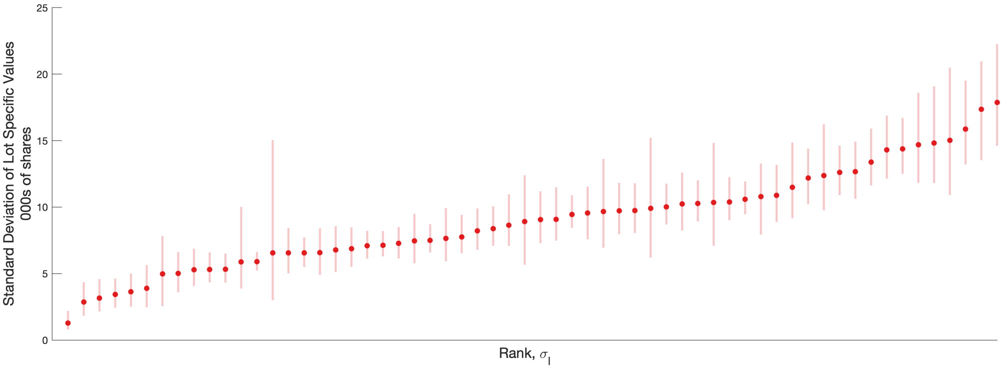
Note: Plot shows estimated marginal value of wealth: The opportunity cost of spending a single share. Results are relative to the food bank with median wealth (normalised to 1). Estimates are sorted across food banks. 95% credible intervals are plotted. return

Results: Estimated Subcategory Weights



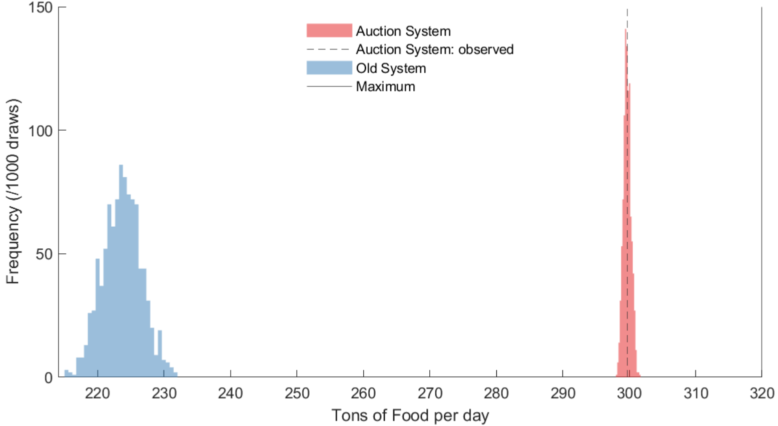
Note: Plot shows the willingness to pay for a ton of food from each of the 152 subcategories of food. Estimates are sorted across subcategories. 95% credible intervals are plotted. return

Results: Estimated Standard Deviation of the Idiosyncratic Value



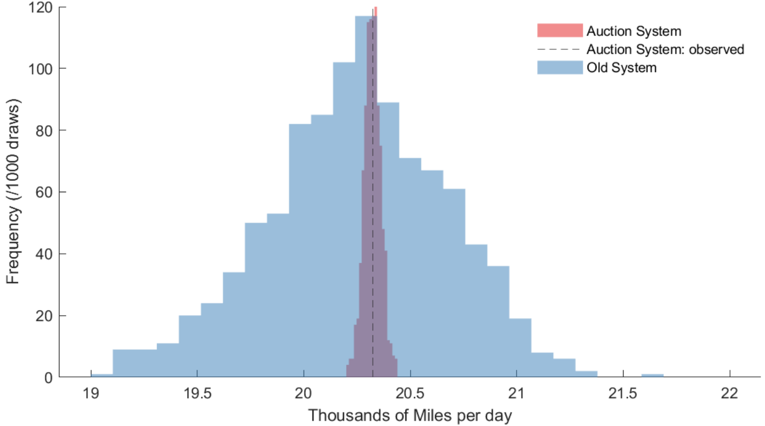
Note: Plot shows the estimated standard deviation parameters of the idiosyncratic lot specific value, essentially the residual. Estimates are sorted across category combinations. 95% credible intervals are plotted.

return



Note: Figure shows the distribution of the quantity of food allocated, weighted by estimated preference weights, measured in equivalent increase in the food supply. Using 1000 draws from the posterior distribution of parameters. On average, the increase allocated food is worth around 15 additional tons of food each day.

return



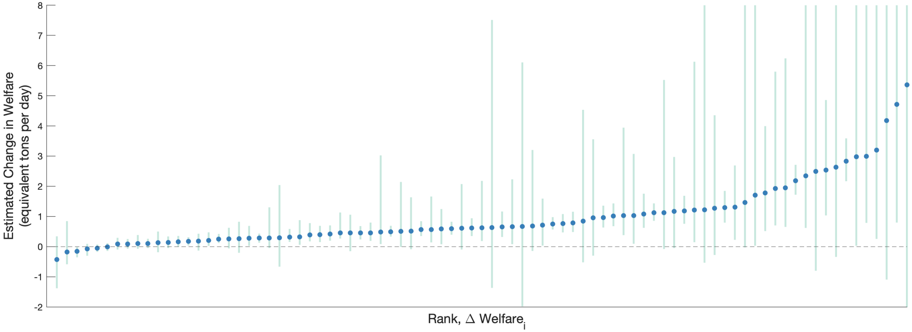
Note: Figure shows the distribution of transportation costs under each of the mechanisms, measured in equivalent increase in the food supply. Using 1000 draws from the posterior distribution of parameters. On average, the reduction in transportation costs is worth around 23 additional tons of food each day.

return

Equity

Welfare by food bank

return



Note: Plot shows estimated welfare by food bank, ordered by posterior mean, with 95% credible intervals.

References